

**MANGALORE UNIVERSITY**  
**B.Sc. PROGRAMME**  
**REVISED SYLLABUS [2024-25 ONWARDS]**  
**COURSE PATTERN AND SCHEME OF EXAMINATION**

**CORE SUBJECT: CHEMISTRY**

**PREAMBLE**

The Under-Graduate Board of Studies (UG BOS) in Chemistry has revised and prepared the syllabi (CBCS based) for the Six Semesters of B.Sc. Chemistry Programme to be implemented by Mangalore University from the year 2024-25. The syllabus includes core subject courses, discipline elective courses and skill enhancement courses carrying a total of 36, four and two credits respectively in the semesters I to VI of the programme.

A detailed skeleton of the entire programme is provided below for the benefit of the aspiring under graduate students.

**Programme Structure for the Bachelor of Science (B.Sc.)**

| Semester    | Course 1     | Course 2     | Course 3     | Optional Course | Language | Compulsory | Total Credit | Total Working hour     |
|-------------|--------------|--------------|--------------|-----------------|----------|------------|--------------|------------------------|
| I           | 5 (3T+2P)    | 5 (3T+2P)    | 5 (3T+2P)    | --              | 3+3      | 2          | 23           | 4+4+4+4+4+4+4+4+2=34   |
| II          | 5 (3T+2P)    | 5 (3T+2P)    | 5 (3T+2P)    | --              | 3+3      | 2          | 23           | 4+4+4+4+4+4+4+4+2=34   |
| III         | 5 (3T+2P)    | 5 (3T+2P)    | 5 (3T+2P)    | 2               | 3+3      | --         | 23           | 4+4+4+4+4+4+4+4+2=34   |
| IV          | 5 (3T+2P)    | 5 (3T+2P)    | 5 (3T+2P)    | 2               | 3+3      | 2          | 25           | 4+4+4+4+4+4+2+4+4+2=36 |
| V           | 8[(2x3T)+2P] | 8[(2x3T)+2P] | 8[(2x3T)+2P] | --              | --       | 2          | 26           | 3+3+4+3+3+4+3+3+4+2=32 |
| VI          | 8[(2x3T)+2P] | 8[(2x3T)+2P] | 8[(2x3T)+2P] | --              | --       | 2          | 26           | 3+3+4+3+3+4+3+4+4+2=32 |
| Grand Total |              |              |              |                 |          |            | 146          | 202                    |

Practical Papers: 2 credits (4 contact hours)

The details of courses offered under the core subject 'Chemistry' is given below. Other important aspects such as basis for internal assessment, pattern of theory question papers and the approximate dates of the internal examinations are also provided.

| Core/ Elective                                                                                                                                                                                            | Course Code        | Title of the Paper                                                         | Instruction Hours | Duration of the Exam (Hrs.) | Max. Marks |    |       | Credits |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------|-------------------|-----------------------------|------------|----|-------|---------|
|                                                                                                                                                                                                           |                    |                                                                            |                   |                             | Exam       | IA | Total |         |
| I Semester B.Sc.                                                                                                                                                                                          |                    |                                                                            |                   |                             |            |    |       |         |
| Group I Chemistry                                                                                                                                                                                         | BSCCHCS101         | Chemistry Theory-I                                                         | 4                 | 3                           | 80         | 20 | 100   | 3       |
|                                                                                                                                                                                                           | BSCCHPS101         | Chemistry Practicals-I Volumetric Analysis                                 | 4                 | 4                           | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject (Chemistry) in I Semester: 05                                                                                                                                    |                    |                                                                            |                   |                             |            |    |       |         |
| II Semester B.Sc.                                                                                                                                                                                         |                    |                                                                            |                   |                             |            |    |       |         |
| Group I Chemistry                                                                                                                                                                                         | BSCCHCS201         | Chemistry Theory-II                                                        | 4                 | 3                           | 80         | 20 | 100   | 3       |
|                                                                                                                                                                                                           | BSCCHPS201         | Chemistry Practicals-II Qualitative Organic Analysis and Chromatography    | 4                 | 4                           | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject (Chemistry) in II Semester: 05                                                                                                                                   |                    |                                                                            |                   |                             |            |    |       |         |
| III Semester B.Sc.                                                                                                                                                                                        |                    |                                                                            |                   |                             |            |    |       |         |
| Group I Chemistry                                                                                                                                                                                         | BSCCHCS301         | Chemistry Theory-III                                                       | 4                 | 3                           | 80         | 20 | 100   | 3       |
|                                                                                                                                                                                                           | BSCCHPS301         | Chemistry Practicals-III Qualitative Analysis of Inorganic Salt mixture    | 4                 | 4                           | 40         | 10 | 50    | 2       |
| Group II Discipline Elective (optional)                                                                                                                                                                   | BSCCHES301         | Laboratory Reagents, Domestic Chemicals and Safety                         | 2                 | 2                           | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject (Chemistry) in III Semester: 05<br>Discipline Elective in III Semester: 02                                                                                       |                    |                                                                            |                   |                             |            |    |       |         |
| IV Semester B.Sc.                                                                                                                                                                                         |                    |                                                                            |                   |                             |            |    |       |         |
| Group I Chemistry                                                                                                                                                                                         | BSCCHCS401         | Chemistry Theory-IV                                                        | 4                 | 3                           | 80         | 20 | 100   | 3       |
|                                                                                                                                                                                                           | BSCCHPS401         | Chemistry Practicals-IV Physical Chemistry Experiments                     | 4                 | 4                           | 40         | 10 | 50    | 2       |
| Group II Discipline Elective (optional)                                                                                                                                                                   | BSCCHES401         | Chemistry in Everyday Life                                                 | 2                 | 2                           | 40         | 10 | 50    | 2       |
| *Compulsory Skill/Practical                                                                                                                                                                               | BSCCHSS401/501/601 | Internship/Mini Project( <b>May be offered in IV or V or VI Semester</b> ) | 2                 | --                          | 40         | 10 | 50    | 2       |
| Total number of Credits for Core Subject (Chemistry) in IV Semester: 05<br>Discipline Elective in IV Semester: 02;<br><br>*Compulsory/Skill/Practical: 02 (May be offered in either IV, V or VI Semester) |                    |                                                                            |                   |                             |            |    |       |         |

| V Semester B.Sc.                                                                                                                                                                             |                        |                                                                                      |   |    |    |    |     |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------|---|----|----|----|-----|---|
| Group I Chemistry                                                                                                                                                                            | BSCCHCS501             | Chemistry Theory-V                                                                   | 3 | 3  | 80 | 20 | 100 | 3 |
|                                                                                                                                                                                              | BSCCHCS502             | Chemistry Theory-VI                                                                  | 3 | 3  | 80 | 20 | 100 | 3 |
|                                                                                                                                                                                              | BSCCHPS501             | Chemistry Practicals-V<br>Gravimetric analysis and<br>Physical Chemistry Experiments | 4 | 4  | 40 | 10 | 50  | 2 |
| *Compulsory Skill/Practical                                                                                                                                                                  | BSCCHSS401/<br>501/601 | Internship/Mini Project<br>(May be offered in IV or V or VI Semester)                | 2 | -- | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject (Chemistry) in V Semester: 08<br>*Compulsory/Skill/Practical: 02 (May be offered in either IV, V or VI Semester)                                    |                        |                                                                                      |   |    |    |    |     |   |
| VI Semester B.Sc.                                                                                                                                                                            |                        |                                                                                      |   |    |    |    |     |   |
| Group I Chemistry                                                                                                                                                                            | BSCCHCS601             | Chemistry Theory-VII                                                                 | 3 | 3  | 80 | 20 | 100 | 3 |
|                                                                                                                                                                                              | BSCCHCS602             | Chemistry Theory-VIII                                                                | 3 | 3  | 80 | 20 | 100 | 3 |
|                                                                                                                                                                                              | BSCCHPS601             | Chemistry Practicals- VI<br>Organic Preparations and Instrumental Methods            | 4 | 4  | 40 | 10 | 50  | 2 |
| *Compulsory Skill/Practical                                                                                                                                                                  | BSCCHSS401/<br>501/601 | Internship/Mini Project<br>(May be offered in IV or V or VI Semester)                | 2 | -- | 40 | 10 | 50  | 2 |
| Total number of Credits for Core Subject (Chemistry) in VI Semester: 08<br>*Compulsory Skill/Practical: 02                                                                                   |                        |                                                                                      |   |    |    |    |     |   |
| Total number of Credits for Core Subject (Chemistry) from I to VI Semester: 36<br>Discipline Elective: 04<br>*Compulsory Skill/Practical: 02 (May be offered in either IV, V or VI Semester) |                        |                                                                                      |   |    |    |    |     |   |

## OBJECTIVES OF THE SYLLABUS

The revised syllabus is designed to provide both theoretical and practical knowledge in the field of chemistry with a special focus on skill enhancement. The syllabus takes into account the requirements of higher education, maintaining the quality of education and student competency level on par with national and international standards. The syllabus is structured to ensure that the students become aware of the practical applications of chemistry knowledge to build careers in the scientific field.

The syllabus aims to enable students to:

- To acquire knowledge and skills in the field of chemistry
- To generate manpower trained in chemistry to meet the need of industry and academia and to pursue higher studies
- To appreciate, understand and use the scientific method in the solving of problems
- To develop the ability to disseminate chemical information effectively
- To acquire good laboratory skills and practice safety measures while handling chemicals

- To understand safe disposal of chemical waste contributing to environmental sustainability
- To apply chemical knowledge to real world situations
- To develop their personality with the necessary skills
- To get good placement

### **Program Outcome**

By the end of the program, the students will be able to

- Understand the applications of chemistry in various fields.
- Get the broad and balanced knowledge of chemistry.
- Develop practical skills which can be applied in actual practice.
- Get the knowledge necessary for employment and higher education.

### **Basis for Internal Assessment, Pattern of Theory Question Papers and Practical Examination**

#### **1. Basis of Internal Assessment in Theory and Practicals**

The internal assessment marks in theory papers shall be based on two tests. The tests shall be at least 1 hour duration each and to be conducted after 6 and 12 weeks after the start of a semester. The average of the two tests shall be taken as the internal assessment marks in theory papers.

The practical internal assessment marks shall be based on one test and continuous evaluation during the practical classes. The practical test shall be conducted after 10 weeks after the start of a semester. The average of the test and continuous evaluation shall be taken as the internal assessment marks in practicals.

#### **1. Theory Question Papers Pattern**

Theory Question Papers shall carry 80 marks. The Question Paper shall consist of Parts A and B, as detailed below:

**Part A:** Part A shall consist of 12 short-answer questions (Question No. 1, sub-parts a to l) covering all units of the syllabus. For I to IV Semesters, 3 questions shall be drawn from each unit ( $3 \times 4 = 12$  questions). For V and VI Semesters, 4 questions shall be drawn from each unit ( $4 \times 3 = 12$  questions). Each question shall carry 2 marks. Students shall answer any 10 questions, making a total of 20 marks ( $10 \times 2 = 20$ ).

**Part B:** For I to IV Semesters, Part B shall consist of eight essay-type questions (Question Nos. 2 to 9), each carrying 15 marks. The questions shall be drawn from all four units, with two questions from each unit. Each question shall have four divisions. Students shall answer one full question from each unit, making a total of four questions ( $4 \times 15 = 60$  marks).

For V and VI Semesters, Part B shall consist of six essay-type questions (Question Nos. 2 to 7), each carrying 20 marks. The questions shall be drawn from all three units, with two questions from each unit. Each question shall have four divisions. Students shall answer one full question from each unit, making a total of three questions ( $3 \times 20 = 60$  marks).

|                                | Unit I  |         | Unit II |         | Unit III |         | Unit IV |         |
|--------------------------------|---------|---------|---------|---------|----------|---------|---------|---------|
| I to IV Semesters              |         |         |         |         |          |         |         |         |
| Q. Nos.<br>(Max.<br>Marks: 15) | 2       | 3       | 4       | 5       | .6       | 7       | 8       | 9       |
| Marks<br>Splitting             | 3+4+4+4 | 3+3+4+5 | 3+4+4+4 | 3+3+4+5 | 3+4+4+4  | 3+3+4+5 | 3+4+4+4 | 3+3+4+5 |
| V and VI Semesters             |         |         |         |         |          |         |         |         |
| Q. Nos.<br>(Max.<br>Marks: 20) | 2       | 3       | 4       | 5       | 6        | 7       | --      | --      |
| Marks<br>Splitting             | 3+5+6+6 | 3+5+5+7 | 3+5+6+6 | 3+5+5+7 | 3+5+6+6  | 3+5+5+7 | --      | --      |

### 3. Scheme and Valuation Procedure for B.Sc. Chemistry Practical Examination

#### First Semester B.Sc. Chemistry Practicals-I

**Duration: 4 Hours**

**Max. Marks: 40 (Practical-30 & Class Record-10)**

#### Valuation Scheme

**The practical examination shall consist of the following:**

##### **Q1. Exercise set for procedure writing**

**10 Marks**

Outline of the procedure including calculations to be written within the first 15 minutes.

Any one of the exercises may be given for this purpose, irrespective of whether the Candidate has carried out experiment or not.

1. Estimation of manganese in pyrolusite by volumetric method.
2. Estimation of glucose using iodine and sodium thiosulphate.
3. Estimation of vitamin-C.
4. Determination of acetic acid in Vinegar using NaOH.
5. Determination of alkali content in antacid tablet using HCl

##### **Q2. Exercise to be set for actual estimation**

**20 Marks**

Any one of the following exercises is to be set for actual estimation. Examiners shall provide the candidates a detailed procedure for the exercise set:

1. Preparation of standard decinormal solution of sodium carbonate and standardization of hydrochloric acid and estimation of sodium hydroxide in solution.
2. Preparation of standard decinormal solution of potassium biphthalate and Standardization of sodium hydroxide solution and estimation of hydrochloric acid in solution.

3. Preparation of standard decinormal solution of oxalic acid and standardization of Potassium permanganate solution and estimation of Mohr's salt in solution.
4. Preparation of standard decinormal solution of ferrousammonium sulphate (Mohr's salt) and standardization of potassium dichromate solution and estimation of ferric chloride in solution.
5. Preparation of standard decinormal solution of potassium dichromate and standardisation of sodium thiosulphate solution and estimation of copper sulphate in solution.
6. Estimation of a mixture of oxalic acid and sulphuric acid in a solution using potassium permanganate solution and standard sodium hydroxide solution.

**Note:** a) AR/GR chemicals should be used for preparing the stock solutions and reagents.

b) At least grade B pipette should be used.

c) The candidates must be provided with 250 ml volumetric flask and 25 ml pipettes.

d) The different volumes (in the range 20-30 ml) of 1N solutions meant for estimation should be pipette out by the examiners in 250 cm<sup>3</sup> volumetric flasks so that no more than 3 candidates in a batch get the same value of 1N solutions distributed.

### 1. Class Records

**10 Marks**

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

#### i) Marks for experiments recorded: 7 Marks

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

#### ii) Marks for neatness: 3 Marks

### 2. Procedure Writing

**10 Marks**

Essential details of procedure = 6 Marks

Tabulation and calculation = 4 Marks

**3. Actual Estimation****20 Marks****i) Titre values: 16 Marks**

| <b>Errors</b>          | <b>Standardization (Marks)</b> | <b>Estimation (Marks)</b> |
|------------------------|--------------------------------|---------------------------|
| $\pm 0.2 \text{ cm}^3$ | 8                              | 8                         |
| $\pm 0.3 \text{ cm}^3$ | 7                              | 7                         |
| $\pm 0.4 \text{ cm}^3$ | 5                              | 5                         |
| $\pm 0.5 \text{ cm}^3$ | 4                              | 4                         |
| $\pm 0.6 \text{ cm}^3$ | 3                              | 3                         |
| Any other value        | 2                              | 2                         |

**NOTE:** Candidates should retain only three titre values on each set of titrations out of which two concordant values are to be considered. All other values must be struck off by the candidates. If a candidate records more than three titre values, first three values are to be considered and extra titre values are to be ignored. In case the candidate records only one titre value, only 50% of the marks are to be awarded in each case. Examiners are requested to bring this to the notice of the candidates. Every burette reading shall be attested by one of the examiners.

**ii) Calculations: 4 Marks**

|                                |         |
|--------------------------------|---------|
| Normality of prepared solution | 1 Mark  |
| Normality of link solution     | 1 Mark  |
| Final step                     | 2 Marks |

**Second Semester B.Sc. Chemistry Practicals-II****Duration: 4 Hours****Max. Marks: 40 (Practicals-30 & Class Record-10)****Valuation Scheme****The practical examination shall consist of the following:****Q1. Exercise set for procedure writing****10 Marks**

Outline of the procedure to be written within first 15 minutes. Any of the exercise in the syllabus under chromatography may be given for this purpose, irrespective of whether the candidate has carried out the experiment or not.

**A. Thin Layer Chromatography**

1. Separation of green leaf pigments.
2. Separation of mixture of dyes using cyclohexane and ethylacetate (8.5:1.5 v/v).

**B. Paper Chromatography: Ascending and Circular**

1. Separation of mixture of phenylalanine and glycine, alanine and aspartic acid, Leucine and glutamic acid; Spray reagent–Ninhydrin.
2. Separation of mixture of D,L-alanine, glycine, L-Leucine using n-butanol-acetic acid-Water (4:1:5); Spray reagent–Ninhydrin.

**C. Column Chromatography:**

1. Separation of Fluorescein and methylene blue.
2. Separation of leaf pigments from spinach leaves.

**Q2. Exercise set for organic analysis****20 Marks**

Any one of the following compounds may be given for analysis:

Resorcinol, Oxalic acid, Urea, Thiourea, Benzoic acid, p-Cresol, p-Toluidine, Chlorobenzene, Bromobenzene, Nitrobenzene, Benzaldehyde, Acetophenone, Benzamide and Aniline.

**1. Class Records****10 Marks**

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

**i) Marks for experiments recorded: 7 Marks**

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

**ii) Marks for neatness: 3 Marks****2. Procedure writing****10 Marks****i) Essential details of procedure: 8 Marks****ii) Calculations: 2 Marks****3. Organic analysis****20 marks**

|                                |                                       |
|--------------------------------|---------------------------------------|
| Preliminary tests:             | 1 Mark                                |
| Physical constant (M.P/B.P):   | 3 Marks (less than 3% error)          |
| Detection of element:          | 4 Marks (tests for N, S and halogens) |
| Solubility Tests:              | 4 Marks                               |
| Reactions of functional group: | 6 Marks (any two)                     |
| Name and structure:            | 2 Marks                               |



### Third Semester B.Sc. Chemistry Practicals–III

Duration: 4 Hours

Max. Marks: 40 (Practicals-30 & Class Record-10)

#### Valuation Scheme

The practical examination shall consist of the following:

##### Q1. Written viva on qualitative analysis

- a) Any one confirmatory test for acid radical: 2 Marks
- b) Complete group analysis of basic radicals of any one group from I to VI: 4 Marks

##### Q2. Exercise set for inorganic qualitative analysis: 24 Marks

- a) Inorganic systematic qualitative analysis of the mixture of two simple salts containing two anions and two cations using semi micro technique.
- b) A simple powdered mixture of inorganic salts containing two anions and two cations is to be prepared on the spot by examiners from simple salts having the following anions and cations.

Anions:  $\text{CO}_3^{2-}$ ,  $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{I}^-$ ,  $\text{NO}_3^-$ ,  $\text{BO}_3^{3-}$ ,  $\text{SO}_4^{2-}$  and  $\text{PO}_4^{3-}$ .

Cations:  $\text{NH}_4^+$ ,  $\text{Cu}^{2+}$ ,  $\text{Pb}^{2+}$ ,  $\text{Ag}^+$ ,  $\text{Bi}^{3+}$ ,  $\text{Cd}^{2+}$ ,  $\text{Co}^{3+}$ ,  $\text{Al}^{3+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Ba}^{2+}$ ,  $\text{Ca}^{2+}$ ,  $\text{Sr}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Na}^+$  and  $\text{K}^+$ .

##### Note:

1. Mixture requiring elimination of phosphate and borate radicals must be avoided (avoid cations such as  $\text{Ba}^{2+}$ ,  $\text{Ca}^{2+}$ ,  $\text{Sr}^{2+}$  and  $\text{Mg}^{2+}$  when borate or phosphate radicals are given).
2. Mixtures of salts which on double decomposition form precipitates insoluble in dilute HCl (like  $\text{BaSO}_4$ ,  $\text{SrSO}_4$ ,  $\text{PbSO}_4$ ) should not be given.
3. Combinations like  $\text{Cl}^-$ ,  $\text{Br}^-$ ,  $\text{I}^-$ ,  $\text{NO}_3^-$  and  $\text{Br}^-$ ;  $\text{NO}_3^-$  and  $\text{I}^-$ ;  $\text{Cl}^-$  and  $\text{Br}^-$ ;  $\text{Cl}^-$  and  $\text{I}^-$ ;  $\text{Cl}^-$  and  $\text{NO}_3^-$ ;  $\text{Br}^-$  and  $\text{I}^-$  must be avoided.
4. The cations should belong to different groups. For example a combination of  $\text{Ca}^{2+}$  and  $\text{Sr}^{2+}$ ;  $\text{Ba}^{2+}$  and  $\text{Ca}^{2+}$ ;  $\text{Ba}^{2+}$  and  $\text{Sr}^{2+}$ ;  $\text{Mg}^{2+}$  and  $\text{Na}^+$ ;  $\text{Na}^+$  and  $\text{K}^+$ ;  $\text{Mg}^{2+}$  and  $\text{Na}^+$ ;  $\text{Al}^{3+}$  and  $\text{Mn}^{2+}$ ;  $\text{Mn}^{2+}$  and  $\text{Zn}^{2+}$ ;  $\text{Bi}^{3+}$  and  $\text{Cd}^{2+}$  must be avoided.
5. AR and GR grade chemicals are used for preparing mixtures.
6. Different mixtures should be prepared and distributed to the candidates (by lots) so that not more than three candidates in a batch get the same mixture.
7. In case of cations, recording of tests are to be done until two cations are detected and confirmed.

##### 1. Class Records

10 Marks

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

**i) Marks for experiments recorded: 7 Marks**

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

**ii) Marks for neatness: 3 Marks**

**2. Written viva on qualitative analysis: 6 Marks**

- a) Any one confirmatory test for acid radical: 2 Marks
- b) Any one basic radical analysis : 4 Marks

**Inorganic qualitative analysis**

The radicals should be reported along with proper chemical tests done systematically.

|                                   |          |
|-----------------------------------|----------|
| Four radicals reported correctly  | 24 Marks |
| Three radicals reported correctly | 18 Marks |
| Two radicals reported correctly   | 12 Marks |
| One radical reported correctly    | 06 Marks |

**Note:**

1. For detecting only the group to which the cations belong, one mark for each correct group should be given.
2. If the acid radical is reported without the confirmatory test, only two marks to be given.
3. If more than four radicals are reported, reduce three marks for each extra radical reported.
4. In case of anions, confirmatory test is expected.
5. In case of cations confirmatory test is expected only in case of  $\text{NH}_4^+$ .
6. Flame test may be considered as one of the preliminary tests and not as a conclusive test for cation.
7. In case of anions, positive tests should be recorded in detail while the essential negative tests may be recorded in brief.

## **Fourth Semester B.Sc. Chemistry Practicals–IV**

**Duration: 4 Hours**

**Max. Marks: 40 (Practicals-30 & Class record-10)**

### **Valuation Scheme**

**The practical examination shall consist of the following:**

#### **Q1. Written Viva**

**6 Marks**

Any one of the exercises prescribed for Practical IV excluding the actual experiment given for the examination may be given for this purpose.

#### **Q2. Any one of the following physical chemistry experiments can be set for the actual experimental work: 24 Marks**

1. Determination of density and surface tension of the given liquid (specific gravity bottle and stalagmometer to be supplied).
2. Determination of density and viscosity of the given liquid (specific gravity bottle and viscometer to be supplied).
3. Determination of molecular mass of the given non-volatile solute by Walker-Lumsden method (molecular mass of the solute should not be more than 140. Electrolyte such as KCl, NaCl,  $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$  also may be included).
4. a) Determination of miscibility temperature of the following systems:
  - i)  $5\text{cm}^3$  phenol +  $4\text{cm}^3$  of water +  $1\text{cm}^3$  of 1% NaCl solution
  - ii)  $5\text{cm}^3$  phenol +  $3\text{cm}^3$  of water +  $2\text{cm}^3$  of 1% NaCl solution
  - iii)  $5\text{cm}^3$  phenol +  $2\text{cm}^3$  of water +  $3\text{cm}^3$  of 1% NaCl solution
  - iv)  $5\text{cm}^3$  phenol +  $1\text{cm}^3$  of water +  $4\text{cm}^3$  of 1% NaCl solution
  - v)  $5\text{cm}^3$  phenol +  $5\text{cm}^3$  of NaCl solution of unknown concentration.b) From the data obtained, find out the unknown concentration of the sodium chloride solution graphically (Pipettes with safety device be provided for piping out phenol).
5. Determination of specific reaction rate for the acid hydrolysis of methyl acetate at a given temperature using 0.5N HCl or 0.5N  $\text{H}_2\text{SO}_4$  (acid to be provided by the examiners).
6. Determination of 1% of toluene in a mixture of toluene + alcohol by refractometry).

#### **A) Class Records**

**10 Marks**

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

**i) Marks for experiments recorded: 7 Marks**

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

**ii) Marks for neatness: 3 Marks****B) Written Viva****6 Marks**

Viva questions should be exclusively from the prescribed practical syllabus.

**C) Physical chemistry experiments: 24 Marks****Experiment 1: Density and Surface tension***Marking of density value*

|                    |             |
|--------------------|-------------|
| Errors $\pm 1\%$ : | 10 Marks    |
| $\pm 2\%$ :        | 8 Marks     |
| $\pm 3\%$ :        | 7 Marks     |
| $\pm 5\%$ :        | 7 Marks     |
| $\pm 10\%$ :       | 3 Marks     |
| Any other value:   | 2 Marks     |
| Calculation :      | 2+2=4 Marks |

*Surface tension values*

|                            |          |
|----------------------------|----------|
| Errors up to $\pm 8\%$ :   | 10 Marks |
| $\pm 8\%$ to $\pm 12\%$ :  | 8 Marks  |
| $\pm 12\%$ to $\pm 15\%$ : | 5 Marks  |
| $\pm 15\%$ to $\pm 25\%$ : | 3 Marks  |
| Any other value :          | 2 Marks  |

**Experiment 2: Density and Viscosity***Marking of density value*

|                    |              |
|--------------------|--------------|
| Errors $\pm 1\%$ : | 10 Marks     |
| $\pm 2\%$ :        | 8 Marks      |
| $\pm 3\%$ :        | 7 Marks      |
| $\pm 5\%$ :        | 7 Marks      |
| $\pm 10\%$ :       | 3 Marks      |
| Any other value:   | 2 Marks      |
| Calculation :      | 2+2= 4 Marks |

*Viscosity Values*

|                            |          |
|----------------------------|----------|
| Errors up to $\pm 5\%$ :   | 10 Marks |
| $\pm 8\%$ to $\pm 7\%$ :   | 8 Marks  |
| $\pm 12\%$ to $\pm 9\%$ :  | 5 Marks  |
| $\pm 15\%$ to $\pm 12\%$ : | 3 Marks  |
| Any other value :          | 2 Marks  |

### Experiment 3: Molecular Mass

|                            |          |
|----------------------------|----------|
| Error up to $\pm 10\%$ :   | 20 Marks |
| $\pm 10\%$ to $\pm 15\%$ : | 18 Marks |
| $\pm 15\%$ to $\pm 20\%$ : | 16 Marks |
| $\pm 20\%$ to $\pm 25\%$ : | 12 Marks |
| $\pm 25\%$ to $\pm 30\%$ : | 8 Marks  |
| Any other value:           | 4 Marks  |
| Calculation :              | 4 Marks  |

**Note:** Candidate shall not retain more than three values in the answer book. Out of which best two values are to be considered for valuation. If candidate records more than three values, first three recorded are to be considered.

### Experiment 4: Miscibility Temperature

- a) 1% NaCl stock solution should be provided by the examiners.
- b) Unknown concentration to be given should be in between 0.3% - 0.7%
- c) Graph-Proper plot (Minimum 4 points): 4 Marks

Any other plot: 2 Marks

Error in concentration (unknown)

|                            |          |
|----------------------------|----------|
| up to $\pm 5\%$ :          | 20 Marks |
| $\pm 5\%$ to $\pm 8\%$ :   | 18 Marks |
| $\pm 8\%$ to $\pm 12\%$ :  | 16 Marks |
| $\pm 12\%$ to $\pm 15\%$ : | 12 Marks |
| $\pm 15\%$ to $\pm 20\%$ : | 8 Marks  |
| Any other value:           | 4 Marks  |

### Experiment 5: Chemical Kinetics

Graph-Proper plot (Minimum 4 points): 5 Marks

Calculation: 5 Marks

Any other plot: 2 marks

|                            |          |
|----------------------------|----------|
| Error up to $\pm 5\%$ :    | 14 Marks |
| $\pm 5\%$ to $\pm 10\%$ :  | 12 Marks |
| $\pm 10\%$ to $\pm 15\%$ : | 10 Marks |
| $\pm 15\%$ to $\pm 20\%$ : | 8 Marks  |
| Any other value:           | 3 Marks  |

### Experiment 6: Analysis of a liquid mixture by Refractometry

a) Graph-Proper plot: 4 Marks

Best straight line graph with at least 4 points: 4 Marks

Less than four points: 2 Marks

b) Percentage composition of the given mixture in the range

$\pm 5\%$  : 20 Marks

$\pm 7\%$  : 16 Marks

$\pm 9\%$  : 12 Marks

$\pm 12\%$  : 8 Marks

Any other values: 4 Marks

Standard binary liquid mixture of known compositions are to be prepared fresh by the candidates.

Unknown is given by the examiners.

**Note:** in all the above experiments if the calculation is wrong no marks to be given for calculation part, but the examiners are required to calculate the values and award the marks as per scheme.

## Fifth Semester B.Sc. Chemistry Practicals–V

Duration: 4 Hours

Max. Marks: 40 (Practicals-30 & Class Record-10)

### Valuation Scheme

The practical examination shall consist of the following:

**Q1. Viva** 5 Marks

**Q2. Gravimetric Exercise** 16 Marks

Examiners shall supply the solution in two 400cm<sup>3</sup> beakers, for each of the candidates, such that the mass of the precipitate will be in the range of 0.2 to 0.3g. A brief outline of the procedure is to be given. The candidates are required to perform two trials each using the given solutions.

One of the following exercises may be set for the gravimetric exercise.

1. Estimation of barium as barium sulphate in barium chloride solution.
2. Estimation of copper as cuprous thiocyanate in copper sulphate solution.
3. Estimation of Ni as Nickel dimethyl glyoximate in nickel ammonium sulphate solution.
4. Estimation of iron as ferric oxide in ferrous ammonium sulphate solution.
5. Gravimetric estimation of chloride / silver as AgCl in NaCl/AgNO<sub>3</sub> solution.
6. Estimation of magnesium as oxinate in magnesium sulphate solution.

**Q3. Colorimetry/Food Adulteration** 9 Marks

One of the following experiments may be set:

- a. To verify Beer-Lambert Law by Job's or Mole-ratio method
- b. Detection of adulterants in any three food stuffs

### Distribution of Marks

**1. Class Records** 10 Marks

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

**i) Marks for experiments recorded: 7 Marks**

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

**ii) Marks for neatness: 3 Marks**

**2. Viva**

**5 Marks**

Viva questions should be exclusively from the prescribed practical syllabus.

**3. Gravimetric Exercise**

**16 Marks**

**i) Marks for mass of precipitate**

|                 |          |
|-----------------|----------|
| ±2%             | 14 Marks |
| ±3%             | 12 Marks |
| ±4%             | 10 Marks |
| +5%             | 8 Marks  |
| ±6%             | 6 Marks  |
| Any other value | 4 Marks  |

**ii) Calculation**

**2 Marks**

**4. Colorimetry/Food Adulteration**

**9 Marks**

**Experiment a) Colorimetry**

**i) Graph- Proper plot (Minimum 4 points): 7 Marks**

**Any other plot: 5 Marks**

**ii) Tabulation and Calculation: 2 Marks**

**Experiment b) Food Adulteration**

Adulterated milk, ghee/butter, sugar/edible oil are to be given

Detecting adulterants in milk: **3 Marks**

Detecting adulterants in ghee/butter: **3 Marks**

Detecting adulterants in sugar/edible oil: **3 Marks**



## Sixth Semester B.Sc. Chemistry Practicals-VI

**Duration: 4 Hours**

**Max. Marks: 40 (Practicals-30 & Class Record-10)**

### Scheme of Valuation

**Q1. Viva to be conducted during practicals**

**5 Marks**

**Q2. Prepare any one of the following organic compound or inorganic complex and present the sample for inspection**

**5 Marks**

- a. Acetanilide
- b. p-Bromoacetanilide
- c. Benzoic acid
- d. Tetra-amine copper (II) sulphate
- e Sodium tri-oxalato ferrate (III)

**Q3. Perform one of the following experiments**

**20 Marks**

- a) To determine the strength of the given acid mixture (acetic acid + hydrochloric acid) conductometrically using standard alkali solution
- b) To determine the equivalent conductance of sodium chloride by conductometric method.
- c) Potentiometric titration of ferrous ammonium sulphate using potassium dichromate as titrant and calculation of the redox potential of  $\text{Fe}^{3+}/\text{Fe}^{2+}$  system on the hydrogen scale.
- d) To determine the dissociation constant of a weak acid by potentiometric method
- e) To determine the concentration of Cupric ions present in a solution using a colorimeter.

### Distribution of Marks

**1. Class Records**

**10 Marks**

The records certified by the teacher in charge and head of the Chemistry Department should be valued by the examiners.

**i) Marks for experiments recorded: 7 Marks**

Minimum ten exercises should be recorded. Marks are to be reduced proportionately for recording lower number of exercises as follows:

| No. of exercises recorded | Marks to be awarded |
|---------------------------|---------------------|
| 10 and above              | 7                   |
| 9                         | 6                   |
| 8                         | 5                   |
| 7                         | 4                   |
| 6                         | 3                   |
| Less than 6               | 0                   |

Repeated recording of the same experiments should not be considered.

**ii) Marks for neatness: 3 Marks**

**2. Viva to be conducted during practicals**

**5 Marks**

Viva questions should be exclusively from the prescribed practical syllabus.

**3. Preparation of organic compounds**

**5 Marks**

i) Marks for preparation of crude sample: 3 Marks

ii) Yield (upto  $\pm 20\%$  of theoretical value): 2 Marks

iii) Any other value: 1 Mark

**4. Instrumental methods**

**20 Marks**

a. To determine the strength of the given acid mixture (acetic acid + hydrochloric acid) conductometrically using standard alkali solution.

i) Graph- Proper plot (Minimum 4 points): 4 Marks

Other plots: 2 Marks

ii) Error in titre values:  $\pm 0.2$  ml 7+7 Marks

$\pm 0.3$  ml 6+6 Marks

$\pm 0.4$  ml 5+5 Marks

$\pm 0.5$  ml 4+4 Marks

Other values 2+2 Marks

iii) Calculation of strength: 2 Marks

b. To determine the equivalent conductance of sodium chloride by conductometric method

i) Cell constant

**5 Marks**

Calculation: 2 Marks

Correct value: 3 Marks

ii) Equivalent conductance

**10 Marks**

Calculation of equivalent conductance values: 6 Marks

Graph-Proper plot: 4 Marks

Other plot: 2 Marks

iii) Correct value of equivalent conductance at infinite dilution:

**5 Marks**

c. Potentiometric titration of ferrous ammonium sulphate using potassium dichromate as titrant and calculation of the redox potential of  $\text{Fe}^{3+}/\text{Fe}^{2+}$  system on the hydrogen scale

i) Graphs- Proper plots 8 Marks (4+4 Marks)

Other plots 4 Marks (2+2 Marks)

ii) Error in titre value  $\pm 0.2$  ml 10 Marks

$\pm 0.3$  ml 8 Marks

$\pm 0.4$  ml 6 Marks

$\pm 0.5$  ml 4 Marks

Other values 2 Marks

|                                                                                                                                                                     |          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| iii. Calculation of redox potential                                                                                                                                 | 2 Marks  |
| d. To determine the dissociation constant of a weak acid by potentiometric method                                                                                   |          |
| i) Graph: Proper plot                                                                                                                                               | 8 Marks  |
| Other plots                                                                                                                                                         | 6 Marks  |
| ii) Error in titre value $\pm 0.2\text{ml}$                                                                                                                         | 10 Marks |
| $\pm 0.3\text{ml}$                                                                                                                                                  | 8 Marks  |
| $\pm 0.4\text{ml}$                                                                                                                                                  | 6 Marks  |
| $\pm 0.5\text{ml}$                                                                                                                                                  | 4 Marks  |
| Other values                                                                                                                                                        | 2 Marks  |
| iii. Calculation of dissociation constant                                                                                                                           | 2 Marks  |
| e. To determine the concentration of Cupric ions present in a solution using a colorimeter (The unknown solution should be in the range of 4 to 6 mM concentration) |          |
| i) Graph-Proper plot (Minimum 4 points):                                                                                                                            | 8 Marks  |
| Other plot                                                                                                                                                          | 6 Marks  |
| ii) Error in concentration $\pm 0.2\text{ mM}$                                                                                                                      | 10 Marks |
| $\pm 0.3\text{ mM}$                                                                                                                                                 | 8 Marks  |
| $\pm 0.4\text{ mM}$                                                                                                                                                 | 6 Marks  |
| $\pm 0.6\text{mM}$                                                                                                                                                  | 4 Marks  |
| Other values                                                                                                                                                        | 2 Marks  |
| iii. Calculation                                                                                                                                                    | 2 Marks  |

## **IV/V/VI Semester B.Sc. Chemistry**

### **Internship/Mini Project**

**Max. Marks: 50 (Report-40 & Internal Assessment-10)**

#### **Valuation Scheme**

**a) Internal Assessment: 10 Marks**

**Basis for Internal Assessment**

Continuous Evaluation: 10 Marks

**b) Evaluation of Report: 40 Marks**

i) Work carried out: 20 Marks

(The Internship/Mini Project may be undertaken either in the Department or in a recognized industry/research institution. During this period, students may be permitted or deputed to visit or work in relevant industries, research laboratories, or institutions to gain practical exposure and hands-on experience.)

ii) Preparation of Report: 10 Marks

iii) Presentation and Viva-Voce: 10 Marks

## **First Semester B.Sc.**

### **BSCCHCS101: Chemistry Theory-I**

**[Total number of Lecture Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Learning Objectives**

**This course helps to understand the following basic aspects of chemistry**

1. Principles of chemical kinetics and theories of reaction rate.
2. Chemical and physical characteristics of solvents.
3. Different aspects of adsorption process.
4. Nature of chemical bonds in molecules.
5. Fundamentals of reaction mechanism.
6. Basic methods of qualitative and quantitative analysis.

#### **Course Outcome**

On completion of this course, the student will be able to appreciate the following aspects.

1. Principles of chemical kinetics and different theories of reaction rate.
2. Adsorption isotherms and adsorption by liquids.
3. Physical and chemical properties of solvents.
4. Nature of bonding in organic molecules and criteria for aromaticity, resonance, hyper conjugation etc.
5. The concepts of organic reactions and techniques of writing the reaction mechanism.
6. Basics of analytical methods and chromatographic techniques.
7. Analytical skills involved in volumetric analysis.

### **UNIT-I**

#### **Chemical Kinetics**

**[5 Hours]**

Concentration dependence of rates, differential rate laws of simple chemical reactions, Zero, First, Second,  $n^{\text{th}}$  order and pseudo first order reactions. Derivation of rate constants for second order and  $n^{\text{th}}$  order reactions with equal initial concentrations. Determination of order of a reaction-Differential, Integration, Half-life period and Isolation methods. Transition state theory-Derivation of relationship between rate constant and equilibrium constant. Thermodynamic parameters of activation.

#### **Surface Chemistry**

**[4 Hours]**

Adsorption of gases on solids: Freundlich and Langmuir adsorption isotherms. Multilayer adsorption-BET equation. Determination of surface area and area of crosssection of a molecule. Adsorption from solution - Gibb's Adsorption isotherm.

**Solvents****[5 Hours]**

Physical properties of a solvent - density, dipole moment, specific conductance, dielectric constant. Types of solvents - classification into protic - aprotic, acidic -basic - amphiprotic, ionizing - non ionizing solvents (examples), Characteristics-liquid range, auto-ionization and solvating properties. Reactions in aqueous and non-aqueous solvents (explanation with examples). Water-hydration, hydrolysis, acid-base, reduction-oxidation, complex formation and precipitation. Ammonia-Ammoniation, ammonolysis, acid-base, reduction-oxidation, complex formation, precipitation, alkali metals in ammonia. Levelling effect of solvents-examples.

**UNIT-II****Chemical Bonding**

Covalent bond-Valence bond theory-Concept of hybridization, Valence Shell Electron Pair Repulsion (VSEPR) theory, Comparative study of structure and bonding between  $F_2O$  and  $H_2O$ ,  $H_2S$  and  $H_2O$ ,  $NH_3$  and  $NF_3$ ,  $ClF_3$  and  $XeOF_2$ . **[7 Hours]**

Basic principle of Molecular Orbital Theory. Molecular orbital diagrams of homo and hetero nuclear species- $N_2$ ,  $O_2$ ,  $CO$ ,  $NO$  and  $CN^-$ . **[4 Hours]**

Ionic bond-Lattice energy, Born-Landé equation, Solvation and Solubility of ionic solids. Polarising power and Polarizability of ions. Fajan's rules to explain bond character, covalent character of ionic compounds, relative covalent character.

Comparative trend in properties: a) Melting point-e.g.  $NaBr$ ,  $MgBr_2$ ,  $AlBr_3$ ;  $LiF$ ,  $LiCl$ ,  $LiBr$ ,  $LiI$ ;  $CaCl_2$ ,  $HgCl_2$ . b) Solubility-e.g.  $AgF$ ,  $AgCl$ ,  $AgBr$ ,  $AgI$  c) Thermal stability-e.g.  $BeCO_3$ ,  $MgCO_3$ ,  $CaCO_3$ ,  $SrCO_3$ ,  $BaCO_3$ . Metallic Bond-Application of Band theory. **[3 Hours]**

**UNIT-III****Nature of Bonding in Organic Molecules****[3 Hours]**

Localised and Delocalised bonds. Conjugation and Cross conjugation. Resonance. Aromaticity-Hückel rule, explanation with examples. Antiaromaticity. Hyperconjugation-relative stabilities of primary, secondary and tertiary carbonations. Electron displacements in covalent bond. Inductive effect and Field effect-Explanation with examples. Concepts of organic acids and bases. Relative strengths of aliphatic and aromatic carboxylic acids-Acetic acid with Chloro acetic acid, Propionic acid and Benzoic acid. Anomalous basic strength of tertiary alkyl amines. Steric effect-Relative stabilities of trans- and cis-2-butene, relative reactivities of alkyl halides in  $S_N2$  reaction, steric hindrance in esterification of acids.

**Mechanism of Organic Reactions****[7 Hours]**

Cleavage and formation of covalent bonds. Notations used to represent electron movements and directions of reactions-arrows, curved arrows, half-headed and double-headed arrows. Types of bond cleavage-homolytic and heterolytic. Substrate and reagent. Types of reagents-Electrophiles and

Nucleophiles-explanation with examples. Types of organic reactions- Substitution, addition, elimination and rearrangement reactions, explanation with examples. Reactive intermediates-Carbo cations, carbanions, free radicals, carbenes, arynes and nitrenes-explanation with examples. Mechanism of Friedel–Craft’s reaction, Cannizzaro reaction, Hofmann rearrangement, addition of HCN and NaHSO<sub>3</sub> to carbonyl compounds (benzaldehyde and acetophenone).

### **Electrophilic Addition to Carbon-Carbon Multiple Bonds**

**[4 Hours]**

Addition of halogens to alkenes-carbocation and halonium ion mechanisms. Stereospecificity of halogen addition. Limitations of  $\pi$  open carbo cation mechanism. Ozonolysis – Mechanism of ozonolysis of propene. Addition of hydrogen halides to alkenes-mechanism, regioselectivity and relative rates of addition. Markownikoff’s and Anti-Markownikoff’s addition of HBr to propene. Hydrogenation, hydration, hydroxylation and epoxidation of alkenes-Explanation with examples. Electrophilic addition to conjugated dienes-mechanism of addition of HBr to 1,3-butadiene, effect of temperature. Free radical addition to 1,3-butadiene. Diels-Alder reaction and its importance, 1,3-Dipolar cycloaddition and Pericyclic reaction-explanation with example.

## **UNIT-IV**

### **Chromatography**

**[3 Hours]**

Chromatographic methods for the separation, concentration and identification of organic compounds-Thin layer, paper and column chromatography. R<sub>f</sub> value and its significance. Principle and applications of Gas Chromatography.

### **Methods of Analysis**

**[7 Hours]**

Qualitative analysis - Sample size and techniques- macro, semi micro and micro.Types of tests- wet, dry and spot tests. Quantitative analysis-Volumetry, Gravimetry and Instrumental analytical methods. Principles of gravimetric analysis-methods of precipitation, optimum conditions for precipitation, co-precipitation and post precipitation. Solvent extraction-basic principles and applications. Errors in quantitative analysis, types of errors- determinate and indeterminate, methods of minimizing errors.Accuracy-absolute error/relative error.Precision–mean deviation /relative mean deviation, standard deviation, t-test, F-test and Q-test. Significant figures. Rules for computation of results (numerical problems to be solved wherever necessary).

### **Periodic Properties**

**[4 Hours]**

Methods of determination of atomic properties -Atomic size by Lande's method, ionization energy by discharge tube method, electron affinity from Born-Haber cycle and electronegativity from Pauling and Mulliken scales. Predicting and explaining the chemical behaviour of elements on the basis of periodic properties (metallic/non-metallic, ionic/covalent, reducing/oxidizing). Effective nuclear charge- shielding effect. Slater's rule and its applications.

## **BSCCHPS101: Chemistry Practicals-I**

### **Volumetric Analysis**

**[Total number of Practical Hours: 4 Hours/Week (4x14=56 Hours)]**

**Objectives:** To understand the concepts and develop the skill of volumetric analysis.

**Course Outcome:** After the completion of the course, the student will develop the skill of analysis by volumetric methods.

1. Microscale experiment-Two buret titration and a pipette titration.
2. Preparation of standard sodium carbonate solution, standardization of hydrochloric acid and estimation of sodium hydroxide in solution.
3. Preparation of standard solution of potassium biphthalate, standardization of sodium hydroxide solution and estimation of hydrochloric acid in solution.
4. Preparation of a standard solution of oxalic acid, standardization of potassium permanganate solution and estimation of Mohr's salt in solution.
5. Preparation of standard ferrous ammonium sulphate solution, standardization of potassium dichromate solution and estimation of ferric chloride in solution.
6. Preparation of standard potassium dichromate solution, standardization of sodium thiosulphate solution and estimation of copper sulphate in solution.
7. Estimation of oxalic acid and sulphuric acid in a mixture using standard potassium permanganate solution and standard sodium hydroxide solution.
8. Estimation of calcium content in limestone as calcium oxalate by permanganometry.
9. Estimation of hardness of water by EDTA method.
10. Estimation of manganese in pyrolusite by volumetric method.
11. Determination of acetic acid in commercial vinegar using NaOH.
12. Determination of alkali content in antacid tablet using HCl.
13. Estimation of glucose using iodine and sodium thiosulphate.
14. Estimation of Vitamin C.

### **Reference Books**

1. A Text Book of Inorganic Chemistry-P. L. Soni. 2013, Sultan Chand and Sons.
2. A Text Book of inorganic Chemistry-B. R. Puri and L. R. Sharma, 2000, Shobanlal Nagin Chand.
3. A text book of Inorganic chemistry-Gurdeep Raj, Krishna Prakashan, 2020.
4. A text book of Inorganic Chemistry-Sathya Prakash, 2001, S. Chand.
5. A Text Book of Quantitative Chemical Analysis-A. I. Vogel, 1989, Longman Group.
6. Physical Chemistry-Samuel Glasstone, 1982, ELBS.



7. A Text Book of Physical Chemistry-P. L. Soni, O. P. Dharmarha and U. N. Dash, 2023, Sultan Chand and Sons, New Delhi.
8. Physical Chemistry-Madan R. L. and Tuli G. D., 2010, Sultan Chand and Sons, New Delhi.
9. A Text Book of Advanced Physical Chemistry-Gurdeep Raj, 2009, Goel, Meerut
10. Organic reaction mechanism-V. K. Ahluwalia and R.K.Parashar, 2011, Narosa, New Delhi.
11. Organic Chemistry-S. M. Mukherji, S. P. Singh and R. K. Kapoor, 2012, New Age International.
12. A Guide Book to Mechanism in Organic Chemistry-Peter Skypes, 2003, Pearson.
13. Instrumental Methods of Chemical Analysis-Willard, Merritt, Dean and Skettle, 2004, CBS Publishers.
14. Instrumental Methods of Chemical Analysis-Gurdeep R. Chatwal and Sham K. Anand, 2011, Himalaya Publishing House.

## **Second Semester B.Sc.**

### **BSCCHCS201: Chemistry Theory-II**

**[Total number of Lecture Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Learning objectives**

**This course helps to understand the following aspects of chemistry**

1. The structure and properties of solids, Liquid Crystals and Gases.
2. General characteristics and properties of s and p block elements.
3. The reaction intermediates of organic reactions and predicting the reaction mechanism.
4. Basic concepts of electrophilic and nucleophilic reactions.
5. Characteristics of chemical compounds of industrial importance.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. Molecular structure of solids and their properties.
2. Different types of liquid crystals and their applications.
3. Thermodynamic properties of gases.
4. Applications of chemicals in daily life.
5. General characteristics and properties of s and p block elements.
6. Organic reaction pathways and writing the reaction mechanism.
7. Basic concepts of electrophilic and nucleophilic substitution reactions.

### **UNIT-I**

#### **Solid State**

**[7 Hours]**

Laws of crystallography: Law of constancy of interfacial angle-explanation taking hexagonal crystal system as an example. Law of symmetry. Elements of symmetry-axis of symmetry, plane of symmetry and centre of symmetry-explanation taking cubic crystal system as an example. Law of rationality of indices. Miller indices-Calculation of Miller indices for different planes in a cubic crystal system. Bravais lattices. X-ray diffraction by crystals. Derivation of Bragg's equation. Determination of crystal structure of NaCl and determination of Avogadro number. Caesium Chloride, Zinc blende structures (numerical problems to be discussed).

#### **Liquid Crystals**

**[2 Hours]**

Explanation, classification with examples-Smectic, nematic, cholesteric, disc shaped and polymeric. Structures of nematic and cholesteric phases- molecular arrangements in nematic and cholesteric liquid crystals. Application of liquid crystals in LCDs and thermal sensing.

#### **Gaseous State**

**[5 Hours]**

Maxwell's distribution of molecular velocities-explanation with graph. Most probable, average and RMS velocities and the relation between them. Qualitative discussion of the collision

number, mean free path and collision diameter. Critical phenomena: P-V isotherms of real gases– Andrews's isotherms of carbon dioxide. Continuity of states-Principles. Isotherms of van der Waal's equation. Relationship between critical constants and Vander Waals constants-Derivation of the expressions for  $a$ ,  $b$ ,  $T_C$ ,  $P_C$  and  $V_C$ , Law of corresponding states-Statement, reduced equation of state-Derivation of the equation.

## UNIT II

### s-Block Elements

[6 Hours]

Hydrogen-position of hydrogen in the periodic table. Hydrides-types, preparation, properties and applications. Structure of NaH and  $\text{BeH}_2$ . Complex hydrides- $\text{LiAlH}_4$ ,  $\text{NaBH}_4$ . Preparation and applications. Comparative study of Li and Be with other members of the same group. Comparative study of lattice energy, enthalpy of formation, enthalpy of hydration and solubilities of alkali metal and alkaline earth metal halides, hydroxides and sulphates. Comparison of standard reduction potentials and reducing properties of alkali metals and alkaline earth metals. Complexation tendencies of alkali metals with crown ether, Cryptates.

### p-Block Elements

[8 Hours]

Comparative study of p-Block elements and their compounds-Comparison between Boron and other members of the group.

**Boranes:** Diborane-Preparation, properties, structure and bonding, chemical evidences for the presence of bridge hydrogen.  $\text{B}_4\text{H}_{10}$ ,  $\text{B}_5\text{H}_9$ -Preparation and structure, Styx number, Wade's rule- Closo, Nido and Arachno boranes. Silicates-Types, basic units, structure and applications. Hydrazine and hydroxylamine-Structure and reducing property. Hypophosphorous acid, phosphorous acid, phosphoric acid, orthophosphoric acid, meta phosphoric acid and pyro phosphoric acid-Structure. Halogens in positive oxidation state. Interhalogen compounds- $\text{ICl}$ ,  $\text{BrF}_3$ ,  $\text{IF}_5$  and  $\text{IF}_7$ -Preparation, properties, structure and uses. Noble gases-Structure and bonding in: Clathrates,  $\text{XeF}_2$ ,  $\text{XeF}_4$ ,  $\text{XeF}_6$  and  $\text{XeO}_3$ .

## UNIT-III

### Reactions Involving Intermediates

[6 Hours]

Generation, stability and mechanism of reactions-

- i) Carbocations-Dienone-phenol rearrangement
- ii) Carbanions-Perkin reaction, Aldol condensation and Claisen condensation
- iii) Free radicals-Sandmeyer's reaction
- iv) Nitrenes-Hofmann rearrangement, Curtius rearrangement
- v) Carbenes-Reimer-Tiemann reaction
- vi) Arynes-Benzyne mechanism for the conversion of Bromobenzene to aniline.

Methods of determination of reaction mechanism-Product analysis, intermediates, isotope effects, kinetic and stereo-chemical studies.

### Nucleophilic Substitution at Saturated Carbon

[2 Hours]

Mechanism of  $\text{S}_{\text{N}}1$  and  $\text{S}_{\text{N}}2$  reactions with suitable examples and energy profile diagrams. Stereochemistry and factors affecting  $\text{S}_{\text{N}}1$  and  $\text{S}_{\text{N}}2$  reactions.

**Elimination Reactions****[2 Hours]**

Mechanism of E1 and E2-Explanation with suitable examples, evidences, orientation and stereochemistry. Hoffmann and Saytzeff rules.

**Aromatic Electrophilic and Nucleophilic Substitutions****[4 Hours]**

Aromatic electrophilic substitution-General mechanism with energy profile diagram. Role of  $\sigma$  and  $\pi$ -complexes. Activating and de-activating substituents, Orienting influence, ortho-para ratio. Nucleophilic aromatic substitution reactions-Addition-elimination and Elimination-addition mechanism.

**UNIT-IV****Industrial Chemistry****[14 Hours]**

**Fuels:** Composition, production and applications of natural gas, water gas, producer gas, LPG and biogas.

**Propellants:** Characteristics and applications.

**Glass:** Raw materials, manufacture-Tank furnace, steps in manufacture and annealing of glass. Types of glasses: Composition and uses of hard, soft, Pyrex, jena, flint, safety, optical, fibre, coloured and Crooke's glasses.

**Cement:** Raw materials, manufacture of cement, mechanism of setting of cement. RCC composition and uses. –

**Ceramics:** Raw materials used in modern ceramics, stages in ceramic making, glazing, applications of porcelain.

**Paints:** Constituents of paints and their functions with examples. Manufacture of white lead and lithopone.

**Refractories:** Characteristics, classification with examples and applications.

**Abrasives:** Natural abrasives, synthetic abrasives, characteristics and applications. Silicon carbide and boron nitride-Structure and production.

**Canesugar:** Outline of production and composition, molasses, its composition.

**Paper:** Production of wood pulp and preparation of paper.

**Chemical fertilizers:** Primary nutrients, different types of fertilizers, importance, production of urea, CAN and superphosphate of lime.

## **BSCCHPS201: Chemistry Practicals-II**

### **Qualitative Organic Analysis and Chromatography**

**[Total number of Practical Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Objectives**

To understand the concepts and develop the skill of qualitative analysis.

#### **Course Outcomes**

After the completion of the course, the student will develop the skill of chromatographic technique and qualitative organic analysis.

**I.** Systematic qualitative analysis of mono and bifunctional organic compounds. Determination of melting point/boiling point. Preparation of suitable solid derivative. Following compounds may be given-Resorcinol, oxalic acid, urea, thiourea, thiophenol, benzoic acid, salicylic acid, phenol, p-cresol, aniline, p-nitroaniline, p-toluidine, benzaldehyde, ethylmethylketone, acetophenone, benzophenone, chlorobenzene, bromobenzene, nitrobenzene and benzamide.

**[8 Weeks]**

**II.** Thin Layer Chromatography: Any two of the following.

**[2 Weeks]**

Determination of R<sub>f</sub> values and identification of organic compounds:

- Separation of green leaf pigments (Spinach leaves may be used),
- Preparation and separation of 2,4-dinitrophenyl hydrazones of acetone, 2-butanone, hexan-2- and 3-one using toluene and light petroleum(40:60, v/v)
- Separation of a mixture of dyes using cyclohexane and ethyl acetate(8.5:1.5, v/v)

**III.** Paper Chromatography: Ascending and Circular. Any two of the following:

**[2 Weeks]**

Determination of R<sub>f</sub> values and identification of organic compounds:

- Separation of a mixture of phenyl alanine and glycine, Alanine and aspartic acid, Leucine and glutamic acid. Spray reagent-ninhydrin.
- Separation of a mixture of D,L-alanine, glycine, and L-Leucine using n-butanol, acetic acid-water (4:1:5, v/v). Spray reagent-ninhydrin.
- Separation of monosaccharides-Mixture of D-galactose and D-fructose using n-butanol:acetone: water (4:5:1, v/v), Spray reagent-Aniline hydrogen phthalate.

**IV.** Column Chromatography

**[2 Weeks]**

- Separation of fluorescein and methylene blue.
- Separation of leaf pigments from spinach leaves.

#### **References**

- Analytical Chemistry-John H. Kennedy, 1997, Saunders College, New York.
- Instrumental Methods of Chemical Analysis-Willard, Merritt, Dean and Skettle, 2004, CBS Publishers.
- Instrumental Methods of Chemical Analysis-Gurdeep R. Chatwal and Sham K. Anand, 2011, Himalaya Publishing House.
- A Text Book of Inorganic Chemistry-P. L. Soni, 2013, Sultan Chand and Sons.
- A Text Book of Inorganic Chemistry-B. R. Puri and L. R. Sharma and K. C. Kalia, 1993, Shoban Lal Nagin Chand.
- A Text Book of Inorganic Chemistry-Gurdeep Raj, 2020, Krishna Prakash.
- A Text Book of Inorganic Chemistry-Sathya Prakash, 2001, Sultan Chand and Sons.

8. Concise Inorganic Chemistry-J. D. Lee, 2022, Wiley, India.
9. Principles of Inorganic Chemistry-B. R. Puri, L. R. Sharma and K. C. Kalia, 2014.
10. Selected Topics in Inorganic Chemistry-R.L.Madan, Malik, G. D. Tuli, 2000, Sulthan Chand and Company.
11. Engineering Chemistry-B. K.Sharma, 2020, Krishna Prakashan.
12. Industrial Chemistry -B. K.Sharma, 2023, Krishna Prakashan.
13. Organic Reaction Mechanism-V. K. Ahuwalia and R. K. Parashar, 2011, Narosa Publishing.
14. Organic Chemistry-S. M. Mukherji, S. P. Singh, R.P. Kapoor and R. Das, 2017, New Age.
15. A Text Book of Qualitative Analysis-A. I. Vogel, 2010, Pearson.

## **Third Semester B.Sc.**

### **BSCCHCS301: Chemistry Theory-III**

**[Total number of Lecture Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Course Learning Objectives**

This course helps to understand the following basic aspects of chemistry:

1. The general characteristics and properties of d and f block elements.
2. The characteristic properties and occurrence of lanthanides and actinides in nature and their uses.
3. Various liquid mixtures and their separation techniques.
4. Chemistry of phenols, ethers, epoxides, structure and reactivity of carbonyl compounds.
5. Concepts of acids and bases with respect to various compounds.
6. Nano-chemistry and the applications nano-materials, food technology and corrosion.
7. Thermodynamics and its applications to physical and chemical systems.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. Distinguish between d and f block elements by studying their general properties.
2. Understand the concept of lanthanides, actinides and their comparisons.
3. Understand the principles of thermodynamics and its applications to physical and chemical systems.
4. Describe the chemistry of phenols, ethers, epoxides and structure and reactivity of carbonyl compounds.
5. Gain the knowledge and skills of preparation of nano-materials.
6. Understand various liquid mixtures and their separation techniques.
7. Understand the basic concept of food technology and corrosion.

### **UNIT-I**

#### **Thermodynamics**

**[14 Hours]**

**First Law of Thermodynamics:** Statement, definition of internal energy and enthalpy. Heat Capacity, heat capacities at constant volume and pressure and their relationship. Joule's law, Joule - Thomson coefficient and inversion temperature. Bond dissociation energy and its calculation from thermo-chemical data. Temperature dependence of enthalpy. Kirchhoff's equation.

**Second Law of Thermodynamics:** Need for the Law, different statements of the Law. Carnot cycle and its efficiency, Carnot theorem. Thermodynamic scale of temperature. Concept of entropy, entropy as a state function, entropy change for an ideal gas as a function of V & T,

entropy as a function of P & T, entropy change in physical changes- fusion, evaporation, sublimation and transition. Entropy changes in mixing of ideal gases. Entropy as a criterion of spontaneity and equilibrium.

**Third Law of thermodynamics:** Significance, unattainability of absolute zero. Gibbs and Helmholtz functions; Gibb's function (G) and Helmholtz function (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity. Variation of G with P, V and T (Illustrative problems to be worked out).

## UNIT-II

### Chemistry of d-Block Elements

[6 Hours]

Introduction, General characteristics of transitional elements, general electronic configuration, stabilities of oxidation states, complexing ability, colour, magnetic property-expression for magnetic moment-spin only formula  $\mu_s$ , calculation of  $\mu_s$  for 3d series elements, Reasons for observed trend, Comparative treatment of 4d, 5d series with their analogues in respect of ionic radii, oxidation states, magnetic behavior and stereochemistry.

### Chemistry of f-Block Elements

[4 Hours]

Introduction, Lanthanide contraction, Causes of lanthanide contraction, Occurrence, Isolation of lanthanides by ion-exchange method, Similarities and comparison between lanthanides and actinides, Separation of neptunium, Plutonium and americium and uranium. Problems on calculation of  $\mu_{S+L}$  for trivalent lanthanide ions.

### Nano Chemistry

[4 Hours]

Introduction, General methods of synthesis, characterization techniques, Scanning Electron Microscopy (SEM)-Principle and method of determination of nanomaterials, advantages over other microscopes, Fullerenes, Preparation of nanoparticle by chemical method, Applications of nanomaterials.

## UNIT-III

### Reactions and Reactivity of Phenols

[4 Hours]

Comparison of acidic properties of phenols with carboxylic acids, alcohols and carbonic acid. Molecular rearrangements-Fries rearrangement, Claisen rearrangement, Synthesis of aryloxy acetic acids.

### Structure and Reactivity of Carbonyl Compounds

[7 Hours]

Structure of carbonyl group, Nucleophilic additions to carbonyl group, relative reactivities of aldehydes and ketones-explanation, Mechanism of reactions involving-Hydride shift-Tischenko reaction, i) C-C bond formation-Bucherer hydantoin synthesis, ii) C=C bond formation-Wittig reaction, iii) C=N bond formation-addition of  $\text{NH}_3$  derivatives, iv) C-O bond formation-Acetal synthesis, Michael addition- $\alpha$ ,  $\beta$ -unsaturated aldehydes and ketones.



**Ethers and Epoxides****[3 Hours]**

Chemical reactions of ethers-Cleavage and auto-oxidation with examples. Zeisel's method. Synthesis of epoxides, Acid and Base catalyzed ring opening of epoxides, Orientation of epoxide ring opening with energy profile diagram.

**UNIT-IV****Acids and Bases****[3 Hours]**

Lewis concepts of acids and bases. Modern concepts of acids and bases. Usanovich concept, Lux-Flood concept. Hard and Soft Acids and Bases (HSAB): Classification of acids and bases as hard and soft. Pearson's HSAB concept and its applications.

**Binary Mixtures****[7 Hours]**

**Liquid-Liquid Mixtures:** Ideal liquid mixtures, Raoult's law. Non-ideal system; Azeotropes: HCl-H<sub>2</sub>O and ethanol-water systems. Partially miscible liquids: Phenol-water, trimethyl amine-water and nicotine-water systems. Lower and upper consolute (critical solution) temperature. Effect of impurity on consolute temperature. Immiscible liquids, steam distillation. Nernst distribution law: Definition and applications (Illustrative problems on steam distillation to be worked out).

**Food Technology****[2 Hours]**

Introduction to food chemistry, Fundamentals of food processing, Fundamentals of milk processing, Food analysis, Food packing technology, Food laws and quality assurance.

**Corrosion Chemistry****[2 Hours]**

**Corrosion:** Introduction, Corrosion-an electrochemical phenomenon. Types of corrosion- Galvanic corrosion, Crevice corrosion, Erosion corrosion, Stress corrosion.

**BSCCHPS301: Chemistry Practicals-III****Qualitative Analysis of Inorganic Salt mixture****[Total number of Practical Hours: 4 Hours/Week (14x4=56 Hours)]**

**Course Learning Objective:** To understand the concepts and develop the skill of inorganic salt analysis.

**Course Outcome:** After the completion of the course, the student will develop the skill to analyse the radicals present in inorganic salt mixtures.

**Semi Micro Qualitative Analysis of Inorganic Salt Mixture**

Systematic qualitative analysis of mixture of two simple inorganic salts (containing two cations and two anions)

**Anions:** CO<sub>3</sub><sup>2-</sup>, HCO<sub>3</sub><sup>-</sup>, SO<sub>3</sub><sup>2-</sup>, S<sup>2-</sup>, Cl<sup>-</sup>, Br<sup>-</sup>, I<sup>-</sup>, NO<sub>3</sub><sup>-</sup>, BO<sub>3</sub><sup>3-</sup>, SO<sub>4</sub><sup>2-</sup> and PO<sub>4</sub><sup>3-</sup>.

**Cations:**  $\text{NH}_4^+$ ,  $\text{Cu}^{2+}$ ,  $\text{Pb}^{2+}$ ,  $\text{Ag}^+$ ,  $\text{Bi}^{3+}$ ,  $\text{Cd}^{2+}$ ,  $\text{Co}^{3+}$ ,  $\text{Ni}^{2+}$ ,  $\text{Al}^{3+}$ ,  $\text{Fe}^{3+}$ ,  $\text{Mn}^{2+}$ ,  $\text{Zn}^{2+}$ ,  $\text{Ba}^{2+}$ ,  $\text{Ca}^{2+}$ ,  $\text{Sr}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Na}^+$  and  $\text{K}^+$ .

**Note:** First experiment should be exclusively used for explaining the basic principles of qualitative inorganic analysis and demonstration.

### Reference Books

1. A Text Book of Inorganic Chemistry-P.L. Soni, 1998, Sultan Chand and Sons.
2. A Text Book of Inorganic Chemistry-Puri and Sharma, 2000, Shobanlal Nagin Chand.
3. Advanced Inorganic Chemistry-Gurudeep Raj, 2008, Goel.
4. A Text Book of Inorganic Chemistry-Sathya Prakash, 2001, Sultan Chand and Sons.
5. Engineering Chemistry-B. K. Sharma, 2001, Krishna Prakashan.
6. Nanomaterials-A. K. Bandyopadhyay, 2008, New Age.
7. Nano Science and Technology-V. S. Muralidharan and A. Subramania, 2009, Ane Books.
8. An Introduction to Metallic Corrosion and its Prevention-Raj Narayan, 1988, Oxford and IBH.
9. Vogel's Textbook of Practical Organic Chemistry (including Qualitative Organic Analysis-A. I. Vogel, 1974, Longman.
10. Organic Chemistry-S. M. Mukherji, S. P. Singh and R. K. Kapoor, 2017, New Age.
11. Food Science and Technology-P. Dev Raj, 2011, New Age.
12. A Text Book of Food and Beverage Management-Sudhir Andrews, 2008, McGraw Hill.

### Discipline Elective (Optional)

#### **BSCCHES301: Laboratory Reagents, Domestic Chemicals and Safety**

**[Total number of Lecture Hours: 2 Hours/Week (2x14=28 Hours)]**

#### **Course Learning Objectives**

This course helps to understand the following basic aspects of chemistry:

- 1) Understand general laboratory protocols and safety practices.
- 2) Assess and minimize laboratory hazards.
- 3) Respond to laboratory emergencies and report incidents.
- 4) Explore the role of serendipity and master reagent preparation.
- 5) Understand the chemistry of common cleansing agents.
- 6) Identify the composition and uses of household items.
- 7) Explore the role of chemicals in cosmetics and personal care products.
- 8) Analyze the function of stain removers and their common types.

**Course Outcomes:** On completion of this course, the student will be able to appreciate the following aspects:

- 1) Apply standard laboratory protocols and safety practices.
- 2) Assess and manage laboratory hazards and emergencies.
- 3) Recognize the role of serendipity in scientific discoveries.
- 4) Prepare and safely handle laboratory reagents and equipment.
- 5) Understand the composition and function of cleansing agents.
- 6) Analyze the chemistry of common household items.
- 7) Evaluate the components and applications of cosmetic products.
- 8) Relate chemical properties to real-life applications in the home.

## UNIT-I

### Laboratory Safety

**[4 Hours]**

Introduction. General laboratory protocols: Basic rules, Good Laboratory Practices. Chemical hazards, safety data sheets, symbols and hazard information, storage procedure, Physical hazards, Health hazards, Reaction hazards. Assessing the risks of hazards. Minimizing the risks of hazards: fume hood, ventilation, fire extinguisher, personal protective equipment's, Preparedness for emergencies from uncontrolled hazards: Importance of reporting incidents, response to common emergencies such as fires, explosions, chemicals spills, chemical exposures, injuries.

### Serendipity

**[3 Hours]**

The role of Chance in making Scientific Discoveries. What is Serendipity-Some Serendipitous Inventions in Science; Gun cotton, Velcro, Plastic, X-rays, Microwave, Super glue, Mauve, Teflon, Saccharin, Stainless steel, Matches. Role of Serendipity in Drug discovery: Inventions in Chemistry that enabled the modern world.

### Laboratory Reagents

**[5 Hours]**

Preparation of laboratory reagents and maintenance of electrodes & common laboratory equipments. Methods of expressing concentrations of solution, Preparation of reagents for qualitative analysis of organic and inorganic compounds. Precaution and safety measures during reagent preparation. Numerical problems related to preparation of solutions.

## UNIT-II

### Domestic Chemicals

**[12 Hours]**

**Cleansing Agents:** Chemical composition of soaps, detergents, dishwashers, drain cleaners, bleaching powder, toothpaste, mouthwash, Stain removers, shampoo and stain removers-Explanation with some common examples. Preparation of soaps, detergents and bleaching powder.

**Domestic Items:** Safety matches, Wax candles, shoe polish, mosquito coils, household germicides and pesticides-their chemical composition. Candle preparation.

**Cosmetics:** Talcum powder, nail polish, thinners, skin care, haircare, lipsticks, sun protection

lotions and creams, eye shadow and eye brow pencils, antiperspirants, perfumes, antiperspirants and deodorants-Explanation with examples. Preparation of perfumes.

### **Reference Books**

1. Chemistryat Home: Exploring the ingredients in everyday products-John Emsley, 2015, Royal Society of Chemistry
2. Chemistry in daily life-Kripal Singh, Third Edition, Eastern Academy Education, 2012, PHI Learning Pvt. Ltd., New Delhi.
3. Chemistry in everyday life-Shardendu Kislaya, 2011, Discovery Publishing House Pvt. Ltd.
4. Laboratory Safety, theory and Practice, First Edition, Editors: Anthony Fuscaldo and others, 1980, Elsevier Publications.
5. Chemical Laboratory Safety and Security: A Guide to Developing Standard Operating Procedures, 2016, National Academies Press-Board on Chemical Sciences and Technology, Division on Earth and Life Studies.
6. Chemistry Laboratory Safety Manual, Indian Institute of Science Education and Research, 2018, IISER, Tirupati.
7. Laboratory Safety Manual, NCBS, 2016, C-CAMP.
8. Practical Chemistry-O. P. Pandey, D. N. Bajpai, S. Giri, 2011, Sultan Chand and Sons.
9. Vogel's Qualitative Inorganic Analysis-G. Svehla, 1979, Longman.
10. Text book of Physical Chemistry-B. R. Puri, L. R. Sharma and M. S. Pathania, 2012, Vishal Publishing Co.
11. Science and serendipity: Famous accidental discoveries, Samira Shackle, Thursday, 2<sup>nd</sup> April 2015, New Humanist.
12. The role of serendipity in drug discovery-Thomas A. Ban, Dialogues in Clinical Neuroscience, 2006, 8(3), 335–344.
13. Five Chemistry Inventions that changed the modern world-The Conversation, June 2, 2015.
14. Serendipity, Luck and Wisdom in Research-Patrick J. Hannan, 2006, iUniverse Inc.

## **Fourth Semester B.Sc.**

### **BSCCHCS401: Chemistry Theory-IV**

**[Total number of Lecture Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Course Learning Objectives**

This course helps to understand the following basic aspects of chemistry:

1. Nomenclature, isomerism, theory of coordination compounds and metal ligand bonding.
2. Relationship between colligative properties and molecular weight of solutes.
3. Chemical equilibrium, effect of pressure, temperature and concentration on chemical equilibrium.
4. Phase equilibrium and applications of phase diagram.
5. Relationship between physical properties and molecular structure.
6. Refractometry, radiation and nuclear chemistry.
7. Reactive methylene compounds, different reagents used for the synthesis of organic compounds and reactions of carboxylic acids.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. Nomenclature, EAN rule, isomerism, theory of coordination compounds and metal-ligand bonding in transitional metal complexes.
2. Relationship between colligative properties and molecular weight of solutes.
3. Relationship between physical properties and molecular structure.
4. Basic terminology of phase equilibrium and chemical equilibrium.
5. Different reagents used for the synthesis of organic compounds and reactions of carboxylic acids and their derivatives.
6. Refractometry, its applications, radiation and nuclear chemistry.

### **UNIT-I**

#### **Solutions, Dilute Solutions and Colligative Properties**

**[8 Hours]**

Methods of expressing concentrations-Activity and Activity coefficients. Colligative properties; Raoult's law of relative lowering of vapour pressure. Osmosis and laws of Osmotic pressure. Elevation in boiling point and depression in freezing point. Thermodynamic derivation of the relation between elevation of boiling point and /depression of freezing point and molecular mass of solute, experimental determination of elevation in boiling point by Walker-Lumsden method and depression in freezing point by Beckmann's method (Illustrative problems to be worked out).

#### **Physical Properties and Molecular Structure**

**[4 Hours]**

Optical activity, polarization (Clausius-Mosotti equation), orientation of dipoles in an electric field, dipole moment, measurement of dipole moment-temperature method and refractivity

method, dipole moment and structure of molecules, magnetic properties-para-magnetism and diamagnetism.

### **Refractometry**

**[2 Hours]**

Introduction-Abbe's Refractometer, applications of Refractometry.

## **UNIT-II**

### **Coordination Compounds**

**[5 Hours]**

Introduction, EAN, Nomenclature, illustration with examples including geometrical and optical isomers, bridging ligands. Isomerism in coordination compounds - ionization isomerism, hydrate isomerism, coordinate isomerism, linkage isomerism. Geometrical isomerism and optical isomerism (Coordination numbers 4 and 6).

### **Metal-Ligand Bonding in Transitional Metal Complexes**

**[9 Hours]**

Postulates of Valence Bond Theory (VBT), Examples for  $sp^3$ ,  $dsp^2$ ,  $dsp^3$ ,  $d^2sp^3$  and  $sp^3d^2$  hybridization- $[Ni(CO)_4]$ ,  $[Ni(CN)_4]^{2-}$ ,  $[Cu(NH_3)_4]^{2+}$ ,  $[Fe(CO)_5]$ ,  $[Fe(CN)_6]^{3-}$ ,  $[Co(NH_3)_6]^{3+}$  and  $[CoF_6]^{3-}$ . Explanation for magnetic properties. Limitations of Valence bond theory.

**Crystal Field Theory:** Important concepts of CFT, Crystal field splitting in octahedral, tetrahedral and square planar complexes, Jahn-Teller distortion and crystal field stabilization energy. Calculation of CFSE, weak and strong field ligands, spectrochemical series, explanation for stability, geometry, magnetic and spectral properties. Factors affecting the crystal field splitting. Limitations of CFT.

## **UNIT-III**

### **Reactive Methylene Compounds**

**[2 Hours]**

Keto-enol tautomerism in ethyl acetoacetate and diethyl malonate, Reactions supporting keto and enol forms. Synthetic applications of reactive methylene compounds: Synthesis of alkyl acetic acids, succinic acids, keto acids,  $\alpha,\beta$ -unsaturated acids (Crotonic acid) and 4-methyl uracil.

### **Reagents and their Synthetic Utility**

**[8 Hours]**

Different reagents used for the synthesis of organic compounds with reaction mechanism: i)  $KMnO_4$ -Oxidation of alkenes to vicinal diols, ii)  $OsO_4$ -Synthesis of cis-1,2-diols, iii) Peracids-Baeyer-Villiger oxidation, iv)  $LiAlH_4$ -Reduction of carbonyl compounds into alcohols, v) N-Bromosuccinimide-Allylic bromination of alkenes, vi)  $H_2O_2$ -Dakin reaction, vii)  $NH_2-NH_2$ -Wolf-Kishner reduction, viii)  $Zn-Hg/HCl$ -Clemmenson reduction, ix)  $CrO_2Cl_2$ -Étard reaction.

**Explanation with an example for commonly used synthetic reagents:**  $O_3$ -Ozonolysis, Periodic acid-Oxidation cleavage of vicinal diols into carbonyl compounds, Lead tetraacetate-Oxidative cleavage of vicinal diamines,  $NaBH_4$ -Reduction of carbonyl compounds into alcohols,  $Na$ /ethyl alcohol-Reduction of ester to alcohol by Bouveault-Blanc reduction.

### **Structure and Reactions of Carboxylic Acids and their Derivatives**

**[4 Hours]**

Structure of carboxylic acid and carboxylate ion, Effect of substituents on the acidity of aliphatic and aromatic carboxylic acids (ortho effect). Reactions of carboxylic acids with mechanism: i) Homologation-Arndt-Eistert reaction, ii) Degradation to alkyl halides-Hunsdiecker reaction, iii) Conversion to primary amines-Curtius reaction, iv) Conversion to haloacids-HVZ reaction. Preparation of derivatives of carboxylic acids-Acid chlorides, amides, esters. Reactions of acid derivatives: Conversion of acid chlorides into aldehydes-Rosenmund's reduction.

## **UNIT-IV**

### **Chemical Equilibrium**

**[4 Hours]**

Derivation of relationship between equilibrium constant and free energy  $\Delta G = -RT \ln K_p$ . Thermodynamic derivation of law of mass action. LeChatelier's principle-Statement and applications-Habers process for the synthesis of ammonia. Van't Hoff's reaction Isotherm and reaction isochore (Van't Hoff equation). (Illustrative problems to be worked out).

### **Phase Equilibrium**

**[7 Hours]**

**Phase Rule:** Statement (mathematical expression) and meaning of the terms. Explanation for the terms phase, component and degrees of freedom with suitable examples for each. Derivation of phase rule from thermodynamic consideration. Explanation of phase equilibrium of one component system (Water and Sulphur system) using phase diagram. Two component system: Classification with examples. Simple eutectic system (lead-silver system)-Phase diagram and explanation, desilverisation of lead (Pattinson's Process). Solid solutions: Compound formation with congruent melting point (Mg-Zn system)-Phase diagram and explanation. Compound formation with incongruent melting point (NaCl + water system)-Phase diagram and explanation. Freezing mixtures (acetone-dry ice).

### **Radiation and Nuclear Chemistry**

**[3 Hours]**

Radiolysis of water (using gamma rays), radiation dosimetry, dosimeters, applications inorganic and organic reactions. Application of radioisotopes in the study of Friedel-Craft's reaction, medicine and soil fertility. Industrial applications.

## **BSCCHPS401: Chemistry Practicals-IV**

**[Total number of Practical Hours: 4 Hours/Week (4x14=56 Hours)]**

### **Course Learning Objectives**

To understand the practical knowledge and skills for the determination of physical properties of organic compounds.

### **Course Outcomes**

After the completion of the course, the student will develop the practical knowledge and skills for the determination of physical properties of organic compounds.

Determination/Study of the following (**Minimum of any 10 experiments to be carried out**)

1. Specific reaction rate for the acid catalyzed hydrolysis of methyl acetate at room temperature using 0.5N HCl or 0.5N H<sub>2</sub>SO<sub>4</sub>.
2. Effect of acid strength on the hydrolysis of an ester.
3. Comparison of the catalytic strengths of HCl and H<sub>2</sub>SO<sub>4</sub> by studying the kinetics of hydrolysis of methyl acetate.
4. Rate of decomposition of potassium iodide by hydrogen peroxide.
5. The distribution of iodine between water and carbon tetrachloride.
6. The distribution of benzoic acid between benzene and water.
7. Preparation of arsenious sulphide sol and comparison of the precipitating powers of mono-, di- and tri-valent ions.
8. Density and viscosity of the given liquid (using specific gravity bottle and viscometer).
9. Percentage composition of a given mixture of glycerol and water by viscometry.
10. Density and surface tension of a liquid using stalagmometer.
11. Composition of binary liquid mixture (Alcohol & toluene) by Refractometry.
12. Percentage of NaCl present in phenol-water system.
13. Molecular weight of a non-volatile solute by Walker - Lumsden method.
14. Critical solution temperature of Phenol-water system.

### Reference Books

1. Selected Topics in inorganic Chemistry-Madan, Malik, Tuli, 2000, Sulthan Chand and Sons.
2. A Text Book of inorganic Chemistry-A. K. De, 2001, New Age international.
3. Engineering Chemistry-B. K. Sharma, 2001, Krishna Publication.
4. A Text Book of Quantitative analysis-A. I. Vogel, 1989, Longman.
5. A Text Book of Inorganic Chemistry-P. L. Soni, 1998, Sultan Chand and Sons.
6. A Text Book of Inorganic Chemistry-Puri and Sharma, 2000, Shobanlal Nagin Chand.
7. A Text Book of Inorganic Chemistry-Gurudeep Raj, 2008, Goel.
8. A Text Book of inorganic Chemistry-Sathya Prakash, 2001, Sultan Chand and Sons.
9. Organic Chemistry-P. Y. Bruise, 2000, Pearson Education.
10. Physical Chemistry-Madan and Tuli, 2001, Sultan Chand and Sons.
11. A Text Book of Advanced Physical Chemistry-Gurudeep Raj, 2001, Goel.
12. A Text Book of Physical Chemistry-B. D. Khosla, 2000, Sultan Chand and Sons.
13. Organic Reaction Mechanism-V. K. Ahluwalia and R. K. Parashar, 2010, Narosa.
14. Organic Chemistry-S. M. Mukherji, S. P. Singh and R. K. Kapoor, 2012, New Age.
15. Concise Co-ordination Chemistry-R. Gopalan and V. Ramalingam, 2008, Vikas.



## **Discipline Elective (optional)**

### **BSCCHES401: Chemistry in Everyday Life**

**[Total number of Lecture Hours: 2 Hours/Week (2x14=28 Hours)]**

#### **Course Learning Objectives**

By the end of these topics, students will be able to:

- 1) Understand the composition and nutritional role of food components.
- 2) Describe food processing and preservation methods.
- 3) Analyze food adulteration and its health impacts.
- 4) Evaluate the use of artificial food colorants and chemical contaminants.
- 5) Recognize the need for renewable energy sources.
- 6) Understand the scientific principles of renewable energy systems.
- 7) Identify and explain the applications of alternative energy technologies.
- 8) Evaluate the benefits and future trends of renewable energy utilization.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

- 1) Understand the composition and functional role of food components.
- 2) Apply knowledge of food processing, preservation, and additives.
- 3) Evaluate food safety issues and health impacts.
- 4) Demonstrate awareness of food laws and safe practices.
- 5) Understand the need for renewable energy sources.
- 6) Explain the scientific principles behind various renewable energies.
- 7) Analyze the working mechanisms and applications of renewable technologies.

### **UNIT-I**

#### **Food Chemistry**

**[6 Hours]**

Food as source of energy and structural material. Components of food-Carbohydrates, Proteins, Oils and Fats. Micronutrients-Vitamins, minerals. Chemical substances used in food preparation-Water, common salt, baking powder, vinegar. Food Processing. Food additives, preservatives and flavours. Explanation with examples for the preservation of food by the use of inhibitors, drying, salting, canning, pickling, smoking, packing and refrigeration. Food safety. Soft drinks-Components. Effects on health.

#### **Food Adulteration**

**[4 Hours]**

Definition, common harmful effects, detection of adulteration, Prevention, Food adulteration act, artificial ripening of fruits-Explanation with examples.

#### **Artificial food colorants**

**[4 Hours]**

Coal tar dyes and non-permitted colours and metallic salts. Analysis of pesticide residues in food.

## UNIT-II

### Chemistry for our future

[14 Hours]

**Alternative Sources of Energy:** Need for the search of renewable sources of energy.

**Solar Energy:** Basic properties of solar energy. Applications of solar energy. Transformation of solar energy. Solar heat collectors. Solar photovoltaic collectors. Applications of solar collectors. Examples. Solar power plant.

**Wind Energy:** Basic properties of wind energy. Applications of wind energy. Transformation of wind energy. Wind turbines. Operative characteristics of wind turbines. Wind power plant. Utilization of wind power. Examples. Trends in wind energy utilization.

**Hydro Power:** Basic properties of water energy. Transformation of water energy. Hydro power plant. Utilization of hydropower. Examples. Trends in hydropower utilization.

**Hydrogen Energy:** Production and applications.

**Ocean Energy:** Principles of ocean thermal energy, conversion system. Principles of wave and tidal energy conversion. Transformation of biomass energy. Applications of biomass.

### Reference Books

1. Food: The chemistry of its components-Tom Coultate, 2002, The Royal Society of Chemistry.
2. Food Science and Technology-Geoffrey Campbelt-Platt, 2017, Wiley Blackwell.
3. Food chemistry-H. K. Chopra and P. S. Panesar, 2010, Narosa Publishing.
4. Chemistry at Home: Exploring the ingredients in everyday products-John Emsley, 2015, Royal Society of Chemistry.
5. Chemistry in daily life-Kirpal Singh, 2012, Eastern Academy Education, PHI Learning Pvt. Ltd, New Delhi.
6. Chemistry in everyday life-Shardendu Kislaya, 2011, Discovery Publishing House Pvt. Ltd.
7. Renewable energy sources and emerging technologies-D. P. Kothari, K. C. Singal and Rakesh Ranjan, 2016, PHI Learning Pvt. Ltd, New Delhi.
8. Solar energy: Fundamentals and applications-H. P. Garg and J. Prakash, 2000, McGraw Hill.
9. Biomass regenerable energy-D. O. Hall and R. P. Overend, 1987, Wiley-Blackwel.
10. Introduction to wind turbine aerodynamics- Alois Peter Schaffarczyk, 2014, Springer.
11. Hydrogen and fuel cells: Fundamentals, technologies and applications-Detlef Stolten, 2010, Wiley-Vest.

## **Compulsory Skill/Practical Paper**

**(May be offered in either IV or V or VI Semester)**

### **BSCCHSS401/501/604: Internship/Mini Project Work**

**[Total number of Project Work Hours: 2 Hours/Week (2x14=28 Hours)]**

Internship/Mini project has to be compulsorily carried out in one of the last three semesters (Semester IV or V or VI) with a workload of 2 hours per week under the supervision of a teacher. Report on the internship/Mini project has to be submitted for evaluation. The Internship/Mini Project may be undertaken either in the Department or in a recognized industry/research institution. During this period, students may be permitted or deputed to visit or work in relevant industries, research laboratories, or institutions to gain practical exposure and hands-on experience.

## **Fifth Semester B.Sc.**

### **BSCCHCS501: Chemistry Theory-V**

**[Total number of Lecture Hours: 3 Hours/Week (3x14=42 Hours)]**

#### **Course Learning Objectives**

This course helps to understand the following aspects of chemistry:

1. Applications of complexes and complex formation in metallurgy, qualitative and gravimetric analysis.
2. Knowledge about magnetic properties, electronic spectra, thermodynamic and kinetic aspects of metal complexes.
3. Types conductance of electrolytes and their measurements.
4. The basic concepts of photochemistry.
5. Basic principles and applications of rotational and vibrational spectroscopy in structural analysis.
6. Basic knowledge of stereochemistry of organic compounds and organic compounds of nitrogen.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. The basic principles of rotational spectroscopy and its applications.
2. Types conductance of electrolytes, their measurements and Kohlrausch law.
3. Understand the basic concepts of photochemistry and photochemical reactions.
4. The knowledge of magnetic properties, electronic spectra, thermodynamic and kinetic aspects of metal complexes.
5. The fundamentals of stereochemical aspects of organic compounds and organic compounds of nitrogen.

### **UNIT-I**

#### **Electrochemistry-I**

**[5 Hours]**

Strong and Weak electrolytes, Specific conductance, Equivalent conductance and its determination – its variation on dilution, Debye-Huckel theory, Debye-Huckel-Onsager's equation for strong electrolytes (no derivation). Transport number, definition, Hittorf's rule, determination of transport number by Hittorf's method using attackable and unattackable electrodes and moving boundary method. Kohlrausch's law and its applications & related problems.

#### **Photochemistry**

**[5 Hours]**

Interaction of radiation with matter, difference between thermal and photochemical processes. Primary and secondary processes of a photochemical reaction, Laws of photochemistry:

Grotthuss - Draper law, Stark - Einstein law (only statement), Jablonski diagram depicting various processes occurring in the excited state, qualitative description of fluorescence, phosphorescence, radiative and non-radiative processes (internal conversion, inter system crossing). Quantum yield- definition, reasons for low and high quantum yield & related problems. Explanation for low and high quantum yield reactions taking combination of  $H_2$  and  $Br_2$  and combination of  $H_2$  and  $Cl_2$  as examples. Photosensitized reactions-energy transfer processes, definition of photosensitization. (e.g.: Photosynthesis in plants, dissociation of  $H_2$ , Isomerization of 2-butene and butadiene), photoinhibition.

### **Rotational Spectroscopy**

**[4 Hours]**

Derivation of equation for moment of inertia of diatomic molecule, Diatomic molecule as rigid rotor; derivation of equation for moment of Inertia of diatomic molecule, energy levels of a rigid rotor, selection rules, determination of bond length, qualitative description of non-rigid rotor, isotope effect and related problems.

## **UNIT-II**

### **Application of Metal Complexes and Complexation**

**[2 Hours]**

Applications of Complexes and Complex Formation in Metallurgy: Ag, Au, Al, Ni extractions, Volumetric analysis: Complexometry, masking, demasking techniques with example, Qualitative Analysis: Test for ferrous and ferric ions, nitrate and ammonium ions, Gravimetric Analysis: Precipitation of nickel, magnesium and aluminum ions.

### **Thermodynamic and Kinetic aspects of Metal Complexes**

**[2 Hours]**

A brief outline of thermodynamic stability of metal complexes. Stepwise formation and overall formation constants and factors affecting the stability of complexes. Substitution reactions of square planar complexes, Trans effect, theories and applications of Trans effect.

### **Magnetic Properties of Transition Metal Complexes**

**[4 Hours]**

Origin of Magnetism, terms used in Magnetochemistry: Magnetic induction, Magnetic flux density, Magnetic moment and Magnetic susceptibility, Magnetic permeability. Lande's calculation of theoretical magnetic moment.  $\mu_{S+L}$ , comparison of magnetic moment  $\mu_S$  and  $\mu_{S+L}$  with experimental value of  $\mu$ . Magnetic behavior of substances-Types of magnetic behavior, Methods of determining magnetic susceptibility-Gouy's method.

### **Vibrational Spectroscopy**

**[6 Hours]**

Molecular vibrations, vibrational degrees of freedom, Hooke's law. Energy levels of a simple harmonic oscillator, selection rules, Instrumentation and measurement of IR spectrum intensity and position of IR bands, determination of force constant and qualitative relation of force constant and bond energies, effect of anharmonic motion, Fingerprint region and functional group region.

### UNIT-III

#### Organic Compounds of Nitrogen

[4 Hours]

Nitroarenes: Reduction in acidic, neutral and alkaline media. Mechanism of nucleophilic substitution in nitroarenes. Amines: Separation of mixture of primary, secondary and tertiary amines (Hinsberg and Hofmann's method).

#### Stereochemistry of Organic Compounds

[10 Hours]

Configurational isomerism-optical, geometrical and conformational. Differences between configuration and conformation.

**Optical Isomerism:** Elements of symmetry, molecular chirality, stereogenic centre, chiral and achiral molecules with two stereogenic centres-Lactic acid and Tartaric acid. Enantiomers-properties, resolution of enantiomers. Diastereomers-definition & examples, threo and erythro diastereomers, meso compounds-definition and examples. Inversion (of sugars) and racemization. Relative and absolute configuration, sequence rules, D&L, R& S systems of nomenclature.

**Geometric Isomerism:** Determination of configuration of geometric isomers. E & Z system of nomenclature, geometric isomerism in oximes and alicyclic compounds. Conformational isomerism-conformational analysis of ethane and 1,2-dichloroethane. Conformations of cyclohexane-Newman projection.

### BSCCHCS502: Chemistry Theory-VI

[Total number of Lecture Hours: 3 Hours/Week (3x14=42 Hours)]

#### Course Learning Objectives

This course helps to understand the following aspects of chemistry:

1. Fundamental's quantum mechanics.
2. Elementary idea of flame photometry and thermo analytical methods.
3. Basic principle of electronic spectra of transition metal complexes.
4. Basic aspects and properties of heterocyclic, bioinorganic and organometallic compounds.

#### Course Outcomes

On completion of this course, the student will be able to appreciate the following aspects:

1. Quantum theory, quantum numbers and postulates of quantum mechanics.
2. Principle, instrumentation and applications of flame photometry and thermoanalytical methods.
3. Knowledge on electronic spectra of transition metal complexes.
4. Basic aspects of preparation and properties of heterocyclic and organometallic compounds.
5. Importance of essential and trace elements in biological processes.

## UNIT-I

### Elementary Quantum Mechanics

[10 Hours]

Black-body radiation, Plank's radiation law, photoelectric effect, Compton effect. De-Broglie hypothesis, Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, physical interpretation of the wave function, postulates of quantum mechanics, particle in a one-dimensional box. Setting up of Schrodinger wave equation for H-atom (no separation of variables or solution), quantum numbers and their importance.

### Flame Photometry

[2 Hours]

General principles, Instrumentation, Interference and applications.

### Thermo Analytical Methods:

2 Hours]

Principles and applications (TG, DTA & DTG).

## UNIT-II

### Electronic Spectra of Transition Metal Complexes

[7 Hours]

Introduction, Russel-Saunders's coupling, microstates, Spectroscopic ground state for  $d^n$  system, Term symbols generated by ligands, Types of electronic spectra, Selection rules for d-d transitions, Relaxation of selection rules, Spectrochemical series. Orgel-energy level diagram for  $d^1$  and  $d^9$  systems, discussion of the electronic spectra of  $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$  and  $[\text{Cu}(\text{H}_2\text{O})_6]^{+2}$  complex ions.

### Organometallic Compounds

[7 Hours]

Definition, nomenclature and classification of organometallic compounds. Preparation, properties, bonding and applications of alkyls and aryls of Li and Hg, mononuclear carbonyls and the nature of bonding in metal carbonyls, evidences in support of back bonding, Industrial applications of organometallic compounds-Hydrogenation of alkenes-Wilkinson's catalyst, Fischer-Tropsch synthesis.

## UNIT-III

### Heterocyclic Chemistry

[10 Hours]

Classification and nomenclature, Molecular orbital pictures and explanation for the aromatic characteristics of pyrrole, furan, pyridine, pyrazole and thiazole. Comparison of aromaticity of pyrrole, furan and pyridine. General methods (any two) of synthesis and reactions of pyrrole, furan, pyridine, pyrazole and thiazole, mechanism of electrophilic substitution in furan and pyrrole. Mechanism of electrophilic and nucleophilic substitution in pyridine, comparison of basicity of pyridine, piperidine and pyrrole, condensed five and six membered heterocycles-explanation with examples. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fischer- Indole synthesis, Skraup quinolone synthesis and Bischler-

Napieralski Isoquinoline synthesis, Mechanism of electrophilic substitution reactions of indole and quinoline.

### **Bioinorganic Chemistry**

**[4 Hours]**

Essential and trace elements in biological processes, biological role of metals- $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Na}^{+}$ ,  $\text{K}^{+}$ ,  $\text{Fe}^{2+}$ ,  $\text{Cu}^{2+}$ ,  $\text{Zn}^{2+}$ . Effect of excess intake of metals, metallo porphyrins with reference to hemoglobin and myoglobin.

## **BSCCHPS501: Chemistry Practicals-V**

### **Gravimetric Analysis and Physical Chemistry Experiments**

**[Total number of Practical Hours: 4 Hours/Week (4x14=56 Hours)]**

#### **Course Learning Objectives**

Apply and practice skills of gravimetric estimations, solvent extraction, steam distillation effluent analysis, stereochemical study of organic compounds etc.

#### **Course Outcomes**

After the completion of the course, the student will develop the practical knowledge and skills for the gravimetric estimations, solvent extraction, steam distillation, effluent analysis, stereochemical study of organic compounds etc.

#### **Minimum of any 10 experiments to be carried out**

#### **Inorganic Gravimetric Exercises**

1. Estimation of barium as barium sulphate in barium chloride solution.
2. Estimation of copper as cuprous thiocyanate in copper sulphate solution.
3. Estimation of Ni as Nickel dimethyl glyoximate in nickel ammonium sulphate solution.
4. Estimation of iron as ferric oxide in ferrous ammonium sulphate solution.
5. Gravimetric estimation of chloride / silver as  $\text{AgCl}$  in  $\text{NaCl}/\text{AgNO}_3$  solution.
6. Estimation of magnesium as oxinate in magnesium sulphate solution.
7. Solvent extraction: Separation and estimation of  $\text{Mg (II)}$  and  $\text{Fe (II)}$  ion.
8. Colorimetry:
  - a. Verification of Beer-Lambert Law by Job's method
  - b. Verification of Beer-Lambert Law by Mole ratio method.
9. Adulteration: Determination of adulteration in food stuffs.
10. Effluent analysis: Analysis of effluent water.
11. Steam Distillation:
  - a. Steam distillation of Naphthalene from its suspension in water.
  - b. Steam distillation of Clove oil from cloves.



c. Separation of o-and p-nitrophenols by steam distillation.

### **Stereochemical Study of Organic Compounds via Models**

12. R and S configuration of optical isomers.

13. E and Z configuration of geometrical isomers

14. Conformational analysis of cyclohexane and substituted cyclohexane.

### **Reference Books**

1. Basic concepts of Analytical Chemistry-S. M. Khopkar, 1993, New Age International.
2. Instrumental methods of Chemical Analysis-B. K. Sharma, 1999, Goel.
3. Instrumental methods of Chemical analysis-Gurudeep R. Chatwal and Sham Anand, 1998, Himalaya.
4. Instrumental methods of Chemical analysis-Willard, Merritt, Dean and Skettle, 2004, CBS.
5. A Text Book of Inorganic Chemistry-Gurudeep Raj, 2008, Goel.
6. A Text Book of inorganic Chemistry-Sathya Prakash, 2001, Sultan Chand and Sons.
7. Concise Inorganic Chemistry-J. D. Lee, 1998, Blackwell Science.
8. Principles of Inorganic Chemistry-Puri, Sharma and Kalia, 2000, Shobanlal Nagin Chand.
9. Selected Topics in Inorganic Chemistry-Madan, Malik, Tuli, 2000, Sultan Chand and Sons.
10. A Text Book of Inorganic Chemistry-A. K. De, 2001, New Age International.
11. A Text Book of Quantitative Analysis-A. I. Vogel, 1989, Longman.
12. Inorganic Polymers-G. R. Chatwal, 1993, Himalaya Publishing House.
13. Theoretical Principles of Inorganic Chemistry-Manku, 2001, Tata McGraw Hills.
14. A Text Book of Inorganic Chemistry-Cotton and Wilkinson, 1992, Wiley Interscience.
15. A Text Book of Inorganic Chemistry-Emeleus and Anderson, 1992, New Age international.
16. Organic Chemistry-P. Y. Bruise, 2000, Pearson Education.
17. Agricultural Chemistry-B. A. Yagodin, 1976, Mir Publishers (Moscow).
18. Physical Chemistry-Madan and Tuli, 2001, Sultan Chand and Sons.
19. A Text Book of Advanced Physical Chemistry-Gurudeep Raj, 2001, Goel.
20. A Text Book of Physical Chemistry-B. D. Khosla, 2000, Sultan Chand and Sons.
21. Fundamentals of Molecular Spectroscopy-C. Banwell and E.M. McCash, 1982, Himalaya.
22. Physical Chemistry-Colin N. Banwell, 1998, Himalaya Publishing.
23. Physical Chemistry-Glasstone, 1982, ELBS.
24. A Text Book of Physical Chemistry-P. L. Soni, O. P. Dharmarha and U. N. Dash, 2011, Sultan Chand and Sons.
25. Organic Spectroscopy-William Kemp, 1991, ELBS.
26. Elementary Organic Spectroscopy-Y. R. Sharma, 2013, Sultan Chand and Sons.
27. Systematic Experiments in Chemistry-Arun Sethi, 2008, New Age Interntionl.

28. Organic Spectroscopy-S. K. Dewan, 2020, CBS.
29. Organic Reaction Mechanism-V. K. Ahluwalia and R. K. Parashar, 2010, Narosa.
30. Organic Chemistry-S. M. Mukherji, S. P. Singh and R. K. Kapoor, 2012, New Age International.
31. Bioorganic, Bioinorganic and Supramolecular Chemistry-P. S. Kalsi, J. P. Kalsi and Ashu Chaudhary, 2007, New Age International.
32. Advanced Physical Chemistry-Gurudeep Raj, 2011, Krishna Prakashan.
33. Bio-Inorganic Chemistry-K. Hussain Reddy, 2007, New Age International.
34. Advanced Inorganic Chemistry-Vol II- S. P. Banerjee, 2015, Books & Allied.
35. Essentials of Nuclear Chemistry-H. J. Arnikar, 2016, Wiley Eastern.
36. Organometallic Chemistry-R. C. Mehrotra and A. Singh, 2000, New age International.
37. Heterocyclic Chemistry-Raj K. Bansal, 2020, New Age International,
38. Medicinal Chemistry-G. R. Chatwal, 2010, Himalaya.

## **Compulsory Skill/Practical Paper**

**(May be offered in either IV or V or VI Semester)**

### **BSCCHSS401/501/601: Internship/Mini Project Work**

**[Total number of Project Work Hours: 2 Hours/Week (2x14=28 Hours)]**

Internship/Mini project has to be compulsorily carried out in one of the last three semesters (Semester IV or V or VI) with a workload of 2 hours per week under the supervision of a teacher. Report on the internship/Mini project has to be submitted for evaluation. The Internship/Mini Project may be undertaken either in the Department or in a recognized industry/research institution. During this period, students may be permitted or deputed to visit or work in relevant industries, research laboratories, or institutions to gain practical exposure and hands-on experience.

## **Sixth Semester B.Sc.**

### **BSCCHCS601: Chemistry Theory-VII**

**[Total number of Lecture Hours: 3 Hours/Week (3x14=42 Hours)]**

#### **Course Learning Objectives**

This course helps to understand the following aspects of chemistry:

1. Polymers, polymerization techniques and applications of polymers.
2. Electrodes, electrochemical cells, fuel cells and applications electrochemical cells.
3. Alkaloids, terpenes, fungicides and herbicides.
4. Fundamentals of composites and mass spectra.

#### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. Types of polymers, polymerization techniques and applications of organic and inorganic polymers.
2. Electrodes, electrochemical cells, fuel cells and applications electrochemical cells.
3. Alkaloids, terpenes, Pesticides, fungicides and herbicides.
4. Fundamentals of composites and mass spectra.

### **UNIT-I**

#### **Electrochemistry II**

**[12 Hours]**

Application of Conductivity Measurements: determination of degree of dissociation, determination of  $K_a$  of acids, determination of solubility product of a sparingly soluble salt and conductometric titrations. Reference electrodes: Calomel electrode, quinhydrone electrode and Ag/AgCl electrode. EMF of a cell and its measurements. Computation of cell EMF. Relation between  $\Delta G$  and  $K$  for a cell reaction, Concentration cell with and without transport, Liquid Junction potential, Application of concentration cells-Determination of valency of ions and solubility product, Potentiometric titrations. Determination of pH using hydrogen, quinhydrone and glass electrodes by potentiometric methods and related problems.

#### **Fuel Cells**

**[2 Hours]**

Importance, Working of Hydrogen-Oxygen fuel cell and Methanol-oxygen fuel cell.

### **UNIT-II**

#### **Organic Polymers**

**[5 Hours]**

Introduction, general classes of synthetic polymers-Addition and condensation with examples, Types of polymerizations (i) Free radical polymerization (ii) Cationic polymerization and (iii) Anionic polymerization of vinyl polymers with one example each, Zeigler-Natta polymerization. Condensation polymers-Phenol formaldehyde resins- Bakelite, Epoxy resins and applications.

Natural rubber-composition. Vulcanization, Synthetic rubbers: Buna-S and SBR-preparation and applications, advantages of synthetic rubbers over natural rubbers.

**Inorganic Polymers** [3 Hours]

Preparation, properties, structure and applications of silicones and fluorocarbons. Production and structural features of borazine, boron nitride and silicon carbide.

**Conducting and Biodegradable Polymers** [4 Hours]

**Conducting Polymers:** Introduction, definition and examples-polyaniline, polyacetylene. Mechanism of conduction. Qualitative treatment of doping, Properties-elasticity with high electrical conductivities, Engineering and biological applications

**Biodegradable Polymers:** Introduction, Structure and Properties, Mechanism of breakdown, Applications and uses.

**Composites** [2 Hours]

Introduction, role of matrix in composites, types of matrix, different matrix materials, reinforcement, classification of composites and applications of composites in industry.

**UNIT-III**

**Alkaloids** [4 Hours]

Classification with examples, General properties formation of salts and exhaustive methylation, physical properties and physiological activity. Structural elucidation of nicotine including synthesis. Structural formulae of atropine, cocaine and hygrine.

**Terpenes** [2 Hours]

Classification with examples, Isolation from plant sources. Structural formulae of geraniol, o-pinene and camphor.

**Pesticides, Fungicides and Herbicides** [3 Hours]

Introduction to the structure and properties of Pesticides: i) Organochlorine compounds-Heptachlor, BHC; ii) Organophosphorus Compounds-Malathion, Parathion; iii) Others-Pyrethrin, Baygon. Herbicides: 2,4-dichlorophenoxy acetic acid. Fungicides: Bordeaux mixture.

**Mass Spectrometry** [5 Hours]

Principle and instrumentation of mass spectrometer. Applications in the determination of molecular mass and isotopic abundance. Nitrogen rule, even electron rule, McLafferty rearrangement. Differentiation between 2-methyl butanal and 3-methylbutanal by McLafferty rearrangement.

## **BSCCHCS602: Chemistry Theory-VIII**

**[Total number of Lecture Hours: 3 Hours/Week (3x14=42 Hours)]**

### **Course Learning Objectives**

This course helps to understand the following aspects of chemistry:

1. Knowledge about theory and instrumentation of colorimetry and spectrophotometry.
2. About principle and applications of different spectroscopic techniques.
3. Elementary idea of retrosynthesis.
4. Mechanism of interconversion, chain lengthening and shortening of carbohydrates.
5. Structure and functions of amino acids, peptides and proteins.
6. Knowledge about drugs and chemotherapeutic agents.

### **Course Outcomes**

On completion of this course, the student will be able to appreciate the following aspects:

1. Theory and instrumentation of colorimetry and spectrophotometry
2. Principle and applications of different spectroscopic techniques
3. Basic concept of retrosynthesis
4. Mechanism of interconversion, chain lengthening and shortening of carbohydrates
5. Structure and functions of amino acids, peptides and proteins
6. Fundamentals of drugs.

## **UNIT-I**

### **Colorimetry and Spectrophotometry**

**[4 Hours]**

Introduction, theory of colorimetry and spectrophotometry. Beer-Lambert's law, Instrumentation and applications of colorimetry and spectrophotometry.

### **Ultraviolet (UV) Absorption Spectroscopy**

**[10 Hours]**

Absorption Laws, Concept of molar absorptivity, energy level, types of electronic excitations, Frank-Condon principle (explanation about red shift and blue shift), presentation and analysis of UV spectra, types of electronic transitions, effect of conjugation. Concept of chromophore and auxochrome. Bathochromic, hypochromic, hyperchromic and hypsochromic shifts. UV spectra of conjugated dienes, dienones and  $\alpha$ ,  $\beta$ -unsaturated carbonyl compounds. Woodward Fieser's rules (Problems to be discussed).

## **UNIT-II**

### **Nuclear Magnetic Resonance (NMR) Spectroscopy**

**[11 Hours]**

Introduction, NMR active nuclei, principle of NMR spectroscopy, instrumentation of PMR spectrometer, origin of spectra, solvents used, scales, chemical shift, shielding and deshielding, number of signals obtained from the sample, position of signals and factors affecting the

chemical shift, spin-spin splitting, spin notation and coupling constants, area of signals, interpretation of PMR spectra of simple organic molecules and its structure such as ethyl bromide, ethanol, 2-propanol, acetaldehyde, acetone, 1,1,2-tribromo ethane, ethyl acetate, aniline, nitrobenzene and benzaldehyde.

### **Photoelectron Spectroscopy**

**[3 Hours]**

Basic principles, valence and core binding energies, shifts in energies due to chemical forces, photoelectron spectra of simple molecules, Koopman's theorem.

## **UNIT-III**

### **Carbohydrates**

**[6 Hours]**

**Monosaccharides:** Interconversions of glucose and fructose, chain lengthening of aldoses (Kiliani-Fischer method), Chain shortening (Ruff degradation). Conversion of glucose and mannose-Epimerisation, Mechanism of osazone formation, Amadori rearrangement, Formation of glycosides, ethers (methyl), esters (acetates). Configuration of glucose and fructose, Determination of ring size of monosaccharides (methylation method), Elucidation of cyclic structure of D(+)-glucose, Mechanism of muta-rotation.

### **Amino Acids, Proteins and Peptides**

**[4 Hours]**

Classification based on functional group, Essential and non-essential amino acids, structure and stereochemistry of amino acids- explanation, Acid-base behaviour, isoelectric point and electrophoresis-explanation. Preparation of  $\alpha$ - amino acids from  $\alpha$  halogenated acids. Strecker synthesis and Gabriel synthesis. Reactions due to  $-\text{CO}_2\text{H}$  and  $-\text{NH}_2$  groups, Action of heat on amino acids.

Structure and nomenclature of di-, tri- and polypeptides, classification of proteins based on chemical composition and molecular shape. Peptide structure determination by end group analysis, denaturation of proteins.

### **Drugs and Chemotherapeutic Agents**

**[2 Hours]**

Classification with examples. Synthesis of antipyrine, Chloramine-T, sulphathiazole and sulphanilamide

### **Retrosynthesis**

**[2 Hours]**

Introduction, general terms, synthons and synthetic equivalents, target molecule, general guidelines for disconnection. Retro analysis and synthesis of benzocaine and 4-methoxy acetophenone.

## **BSCCHPS601: Chemistry Practicals-VI**

### **Organic Preparations and Instrumental Methods**

**[Total number of Practical Hours: 4 Hours/Week (4x14 =56 Hours)]**

#### **Course Learning Objectives**

To understand the practical knowledge and skills for the instrumental experiments, preparation of organic and complex compounds.

#### **Course Outcome**

Students should be known how to select a solvent for crystallization, how crystallization should be carried out, complex compound preparation, and conduct experiments using modern instruments.

#### **Minimum of any 10 experiments to be carried out**

#### **Organic Preparations (Preparation, purification and characterization by recording melting point)**

1. Preparation of acetanilide from aniline.
2. Preparation of p-bromoacetanilide
3. Nitration of acetanilide to p-nitro acetanilide and hydrolysis to p-nitroaniline.
4. Preparation of iodoform from ethanol
5. Preparation of m-dinitrobenzene
6. Preparation of adipic acid from cyclohexanol.
7. Preparation of benzoic acid from toluene/benzaldehyde.
8. Preparation of tribromoaniline from aniline.

#### **Instrumental Methods**

9. To determine the strength of the given acid mixture (acetic acid + hydrochloric acid) conductometrically using standard alkali solution.
10. To determine the dissociation constant of a weak acid by potentiometric method.
11. To determine equivalent conductance of sodium chloride by conductometric method.
12. To determine the ionization constant of a weak acid conductometrically.
13. Potentiometric titration of ferrous ammonium sulphate using potassium dichromate as titrant and calculation of the redox potential of  $\text{Fe}^{3+}/\text{Fe}^{2+}$  system on the hydrogen scale.
14. To study the rate of inversion of cane sugar.
15. To determine the concentration of cupric ions, present in a solution using a colorimeter.

#### **Preparation of Complexes**

16. Preparation of sodium trisoxalateferrate (III),  $\text{Na}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ .
17. Preparation of tetra ammine copper (II) sulphate,  $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ .
18. Preparation of hexaminecobalt (III) chloride,  $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ .

**Note:** Principles of physical chemistry experiments, organic preparation and complex preparation are to be discussed in the laboratory.

### Reference Books

1. Instrumental Methods of Chemical Analysis-Gurudeep R. Chatwal and Sham Anand, 1998, Himalaya Publishing House.
2. Instrumental Methods of Chemical Analysis-Willard, Merritt, Dean and Skettles, 2004, CBS.
3. A Text Book of Inorganic Chemistry-Cotton and Wilkinson, 1992, Wiley Interscience.
4. A Text Book of Inorganic Chemistry-P. L. Soni, 1998, Sultan Chand and Sons.
5. A Text Book of Inorganic Chemistry-Puri and Sharma, 2000, Shobanlal Nagin Chand.
6. A Text Book of Inorganic Chemistry-Gurudeep Raj, 2008, Goel.
7. A Text Book of inorganic Chemistry-Sathya Prakash, 2001, S. Chand & Company.
8. Concise inorganic Chemistry-J. D. Lee, 1998, Blackwell Science.
9. Principles of Inorganic Chemistry-Puri, Sharma and Kalia, 2000, Shobanlal Nagin Chand.
10. Selected Topics in inorganic Chemistry-Madan, Malik, Tuli, 2000, Sultan Chand and Sons.
11. A Text Book of Inorganic Chemistry-A. K. De, 2001, New Age international.
12. Industrial Chemistry-B.K. Sharma, 2001, Goel.
13. A Text Book of Quantitative Analysis-A. I. Vogel, 1989, Longman.
14. Inorganic Polymers-G. R. Chatwal, 1993, Himalaya Publishing House.
15. Theoretical Principles of Inorganic Chemistry-Manku, 2001, Tata McGraw Hills.
16. Advanced Physical Chemistry-Gurudeep Raj, 2011, Krishna Prakashan.
17. Agricultural Chemistry by B.A. Yagodin, 1976, Mir Publishers (Moscow).
18. Physical Chemistry by Madan and Tuli, 2001, S. Chand & Company.
19. A Text Book of Advanced Physical Chemistry by Gurudeep Raj, 2001, Goel.
20. A Text Book of Physical Chemistry-P. L. Soni, O. P. Dharmarha and U. N. Dash, 2011, Sultan Chand and Sons.
21. Organic Spectroscopy-William Kemp, 1991, ELBS.
22. Elementary Organic Spectroscopy-Y. R. Sharma, 2013, Sultan Chand and Sons.
23. Systematic Experiments in Chemistry-Arun Sethi, 2008, New Age International.
24. Organic spectroscopy- S. K. Dewan, 2020, CBS.
25. Organic Reaction Mechanism-V. K. Ahluwalia and R. K. Parashar, 2010, Narosa.
26. Organic Chemistry-S. M. Mukherji, S. P. Singh and R. K. Kapoor, 2012, New Age International.
27. A Hand Book of Sustainable Polymers-Vijay Kumar Thakur and Manju Kumari Thakur, 2016, CRC Press.
28. Degradable polymers, Principles and Applications-Gerald Scott, 2002, Kluwer Academic.



29. Hand Book of Biopolymers-Shakeel Ahmed, Suvardhan Kanchi, Gopalakrishnan Kumar, 2018, CRC Press.

## **Compulsory Skill/Practical Paper**

**(May be offered in either IV or V or VI Semester)**

### **BSCCHSS401/501/601: Internship/Mini Project Work**

**[Total number of ProjectWork Hours: 2 Hours/Week (2x14=28 Hours)]**

Internship/Mini project has to be compulsorily carried out in one of the last three semesters (Semester IV or V or VI) with a workload of 2 hours per week under the supervision of a teacher. Report on the internship/Mini project has to be submitted for evaluation. The Internship/Mini Project may be undertaken either in the Department or in a recognized industry/research institution. During this period, students may be permitted or deputed to visit or work in relevant industries, research laboratories, or institutions to gain practical exposure and hands-on experience.

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